

MAGNETOM

# Function Description System

Gradient

## Document Version

Siemens reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order a current version or refer to our website <http://www.healthcare.siemens.com>.

## Disclaimer

Siemens provides this documentation “as is” without the assumption of any liability under any theory of law.

The content described herein shall be used by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or its affiliates to use such documents.

Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are not entitled to use this documentation without prior written authority.

## Copyright

“© Siemens, 2006” refers to the copyright of a Siemens entity such as:

Siemens Healthcare GmbH - Germany

Siemens Aktiengesellschaft - Germany

Siemens Shenzhen Magnetic Resonance Ltd. - China

Siemens Shanghai Medical Equipment Ltd. - China

Siemens Healthcare Private Ltd. - India

Siemens Medical Solutions USA Inc. - USA

Siemens Healthcare Diagnostics Inc. - USA and/or

Siemens Healthcare Diagnostics Products GmbH - Germany

|          |                                          |           |
|----------|------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                      | <b>5</b>  |
| 1.1      | Gradient performance                     | 6         |
| 1.2      | The hardware components                  | 7         |
| 1.2.1    | Function                                 | 10        |
| 1.2.2    | Specifications                           | 10        |
| <b>2</b> | <b>Gradient Small Signal Unit (GSSU)</b> | <b>12</b> |
| 2.1      | Overview                                 | 13        |
| 2.2      | Backplane D18                            | 14        |
| 2.2.1    | Function                                 | 14        |
| 2.3      | DAC Board D11 or D112                    | 15        |
| 2.3.1    | General                                  | 16        |
| 2.3.2    | Function                                 | 16        |
| 2.4      | Regulator Board D12                      | 19        |
| 2.4.1    | General                                  | 19        |
| 2.4.2    | Function                                 | 20        |
| 2.5      | Modulator Board D14                      | 28        |
| 2.5.1    | General                                  | 28        |
| 2.5.2    | Function                                 | 29        |
| 2.6      | Service/CAN Board D16 K2217              | 34        |
| 2.6.1    | Function                                 | 36        |
| 2.6.2    | Switches                                 | 36        |
| 2.6.3    | LEDs                                     | 36        |
| 2.7      | I/O Board D17 K2217                      | 38        |
| 2.7.1    | General                                  | 38        |
| 2.7.2    | Function                                 | 39        |
| 2.8      | CAN & Service Board D139                 | 40        |
| 2.8.1    | General                                  | 40        |
| 2.8.2    | Function                                 | 41        |
| 2.8.3    | Digital Signal Multiplexer (Loop)        | 41        |
| <b>3</b> | <b>GSSU Power Supply</b>                 | <b>42</b> |
| 3.1      | Overview                                 | 42        |
| 3.2      | DC Supply for GSSU                       | 43        |
| 3.2.1    | Function                                 | 43        |
| 3.3      | DC Supply for the Driver Supply          | 44        |
| 3.3.1    | Function                                 | 44        |
| <b>4</b> | <b>Power Stage Supply</b>                | <b>45</b> |
| 4.1      | Overview                                 | 46        |
| 4.1.1    | General                                  | 46        |
| 4.2      | Soft Start Unit D31 K2217                | 47        |
| 4.2.1    | Function                                 | 48        |

|          |                                      |           |
|----------|--------------------------------------|-----------|
| <b>5</b> | <b>Power Stages</b>                  | <b>49</b> |
| 5.1      | Overview .....                       | 50        |
| 5.2      | Power Stage Module .....             | 51        |
| 5.2.1    | Connectors .....                     | 52        |
| 5.2.2    | LEDs .....                           | 52        |
| 5.3      | Cascade Power Stage Switching .....  | 53        |
| 5.3.1    | Function .....                       | 53        |
| <b>6</b> | <b>Current Sensor Assembly (CSA)</b> | <b>54</b> |
| 6.1      | Overview .....                       | 55        |
| <b>7</b> | <b>Gradient Supervision</b>          | <b>56</b> |
| 7.1      | Overview .....                       | 56        |
| 7.2      | Function .....                       | 57        |
| 7.2.1    | Power Supply .....                   | 59        |
| 7.2.2    | Device Calibration .....             | 59        |
| 7.2.3    | Device Fault .....                   | 59        |
| 7.2.4    | Fire Alarm .....                     | 60        |
| 7.3      | Jumpers .....                        | 62        |
| 7.4      | Switches .....                       | 63        |
| 7.5      | LEDs .....                           | 64        |
| 7.6      | Service Procedures .....             | 65        |
| <b>8</b> | <b>Changes to previous version</b>   | <b>66</b> |
| <b>9</b> | <b>List of Hazard IDs</b>            | <b>67</b> |

The Symphony Tim Upgrade systems use the existing K2217 Quantum Gradient Power Amplifier, together with the AS39T Gradient Coil. It is based on the principle of the switch-mode current generators (amplifiers) and actively shielded gradient coils.

To adapt the unit to the new Advanced Measurement Control (AMC), an additional board - the "D139 CAN & Service Board" - needs to be fitted into the free slot of the Gradient Small Signal Unit (GSSU).

## 1.1 Gradient performance

The performance and power of the gradient system is described by different parameters:

- maximum gradient amplitude  $\text{GRAD}_{\text{max.}}$  (e.g.  $20 \text{ mT/m}$ ),
- effective gradient amplitude  $\text{GRAD}_{\text{eff.}} = \text{GRAD}_{\text{max.}} \cdot \sqrt{3}$
- minimum rise time to max. gradient amplitude (e.g.  $400 \text{ } \mu\text{s}$ ),
- max. voltage at the gradient coil (e.g.  $2000 \text{ V}$ ),
- the so-called slew rate SR.

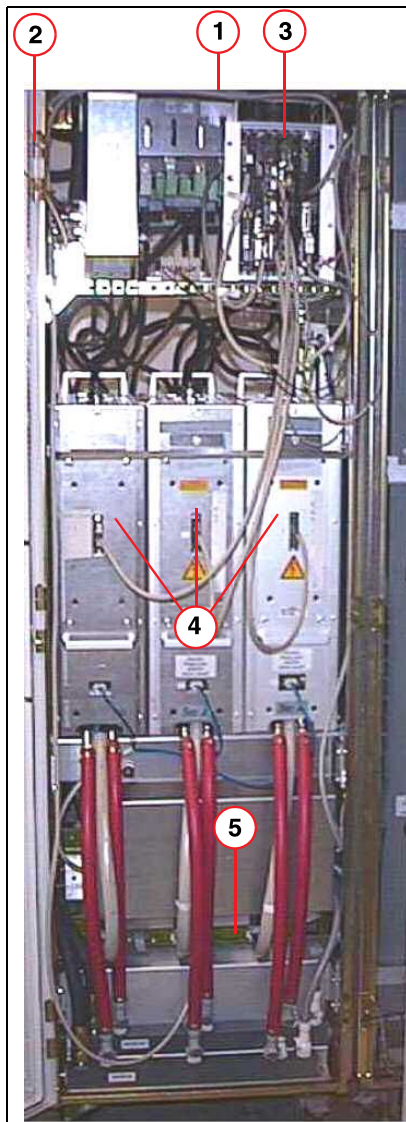
The SR value is calculated from the max. gradient field and the minimum time needed to reach this field (rise time) by the formula:

Fig. 1:

$$SR = \frac{\text{max. gradient field}}{\text{min. rise time}} \quad \text{e. g.:} \quad SR = \frac{20 \frac{\text{mT}}{\text{m}}}{0.400 \text{ ms}} = 50 \frac{\text{mT}}{\text{ms}} = 50 \frac{\text{T}}{\text{s}}$$

## 1.2 The hardware components

Fig. 2: Gradient System K2217



- (1) Current Sensor Assembly (CSA)
- (2) GSSU Power Supply
- (3) GSSU
- (4) Power stages
- (5) Power Stage Supply

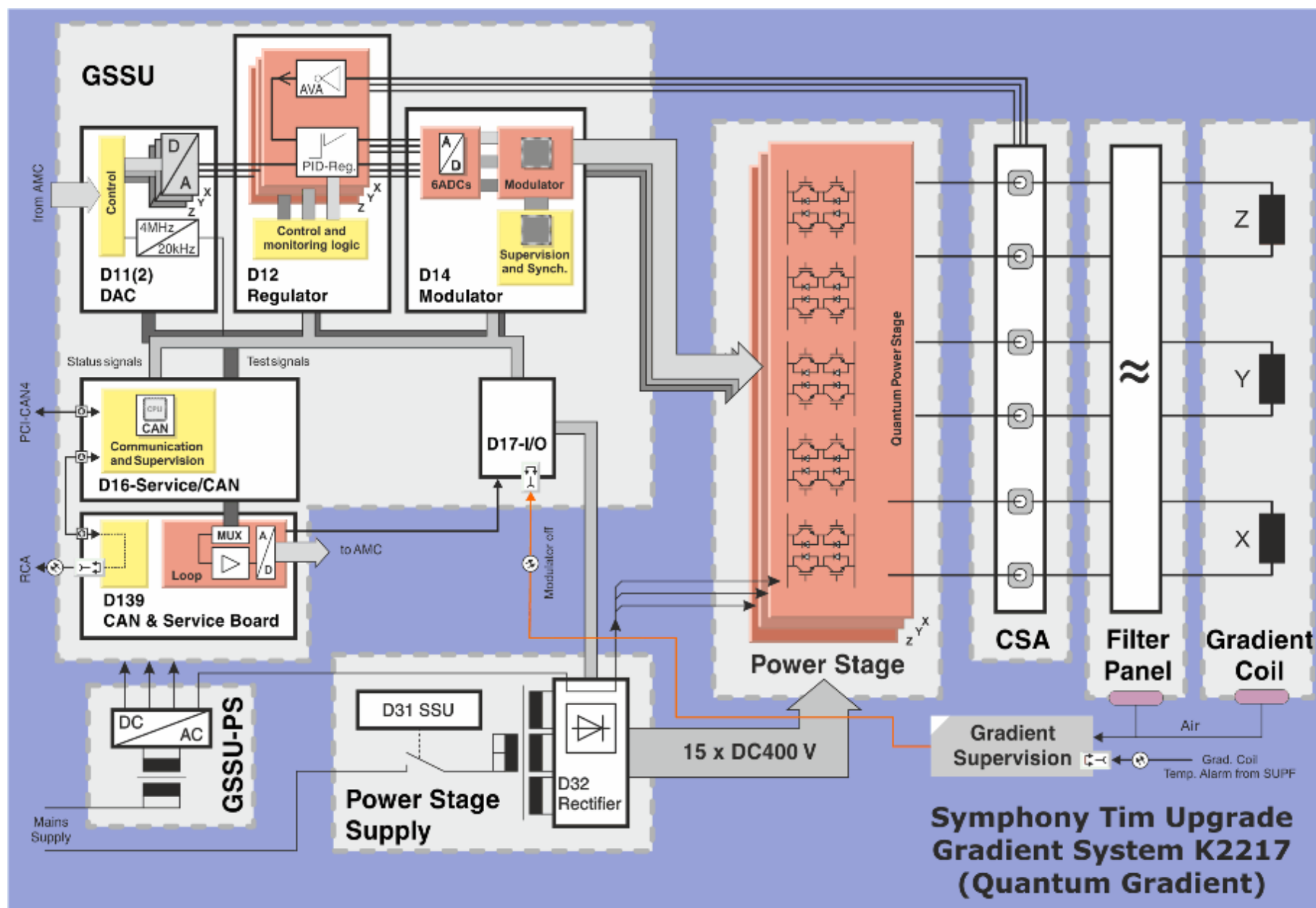
The gradient system hardware components consist of six functional blocks:

- **Gradient Small Signal Unit (GSSU)**
- **GSSU Power Supply**
- **Power Stage Supply**
- **Power Stage (3x)**
- **Current Sensor Assembly (CSA)**

- Gradient Coil Assembly



Fig. 3: Block Diagram of Gradient System K2217



## 1.2.1 Function

### 1.2.1.1 Gradient Small Signal Unit (GSSU)

The task of the GSSU is to control the function of the Gradient Power Amplifier (GPA) generating signals required to drive a precise current in the gradient coils. Additionally, the GSSU provides links to other components of the MR system.

### 1.2.1.2 GSSU Power Supply

The GSSU Power Supply provides stabilized DC voltages ( +5 V and  $\pm 15$  V) for the GSSU and 16.5 VDC for an alternating (25 kHz) voltage inside the Power Stages.

### 1.2.1.3 Power Stage Supply

The Power Stage supply consists of a dedicated transformer and rectifier circuits for the five cascades of the Power Stage. Each cascade is supplied with 400 VDC.

### 1.2.1.4 Power stages

The Power Stage is a switched-mode current generator based on power IGBT arrays. In order to allow a compact construction, the power devices are mounted on water cooled heat sinks. One Power Stage consists of all four switches required to generate bipolar output current with the unipolar power supply.

### 1.2.1.5 Current Sensor Assembly (CSA)

The current sensors are responsible for generating signals proportional to the actual current driven through the gradient coils. In order to provide the required accuracy and the galvanic decoupling from the output current, special devices are used.

### 1.2.1.6 Gradient coil assembly

The gradient coil assembly consists of the active-shielded water-cooled gradient coils and components for the passive and active shim system. The mechanical construction of the gradient coil assembly allows the installation of the coil in different magnet types and supports fixing RF-coils mounted inside of the gradient coil.

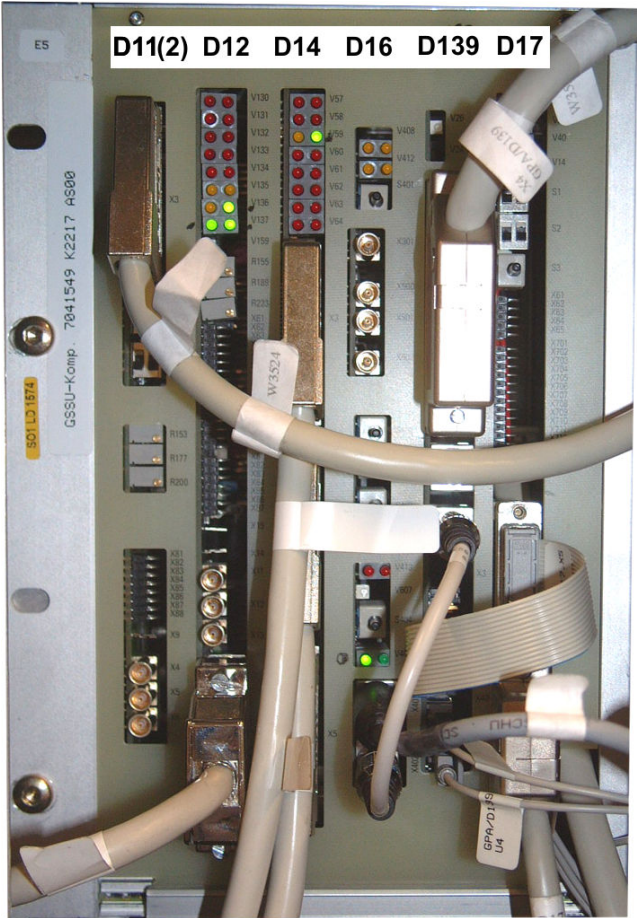
## 1.2.2 Specifications

Tab. 1 Gradient performance

| Parameter                                             | Performance value |
|-------------------------------------------------------|-------------------|
| Max. gradient field (X, Y, Z)                         | 30 mT/m           |
| Min. rise time                                        | 300 $\mu$ s       |
| Slew rate                                             | 125 T/m/s         |
| Max. gradient current (for each of the gradient axes) | 380 A             |

| Parameter                             | Performance value |
|---------------------------------------|-------------------|
| Max. voltage across the gradient coil | 2000 V            |
| Active shielded gradient coil         | AS 39T            |

Fig. 4: GSSU front view



## 2.1 Overview

The GSSU generates the pulse-width modulated control signals for the switches in the power stages based on the digital data coming from the AMC. Its additional task is the overall supervision of the GPA.

The PCBs or FRUs (field replaceable units) of the GSSU that are connected to the one common backplane (D18) are listed below:

- DAC D11 (or DAC D112 on newer GPAs)
- Regulator D12
- Modulator D14
- Service/CAN D16
- I/O board D17
- CAN & Service board D139



On newer GPAs, the DAC D11 was replaced by a DAC D112 board (fully compatible). The D11 description is valid for the D112 as well.

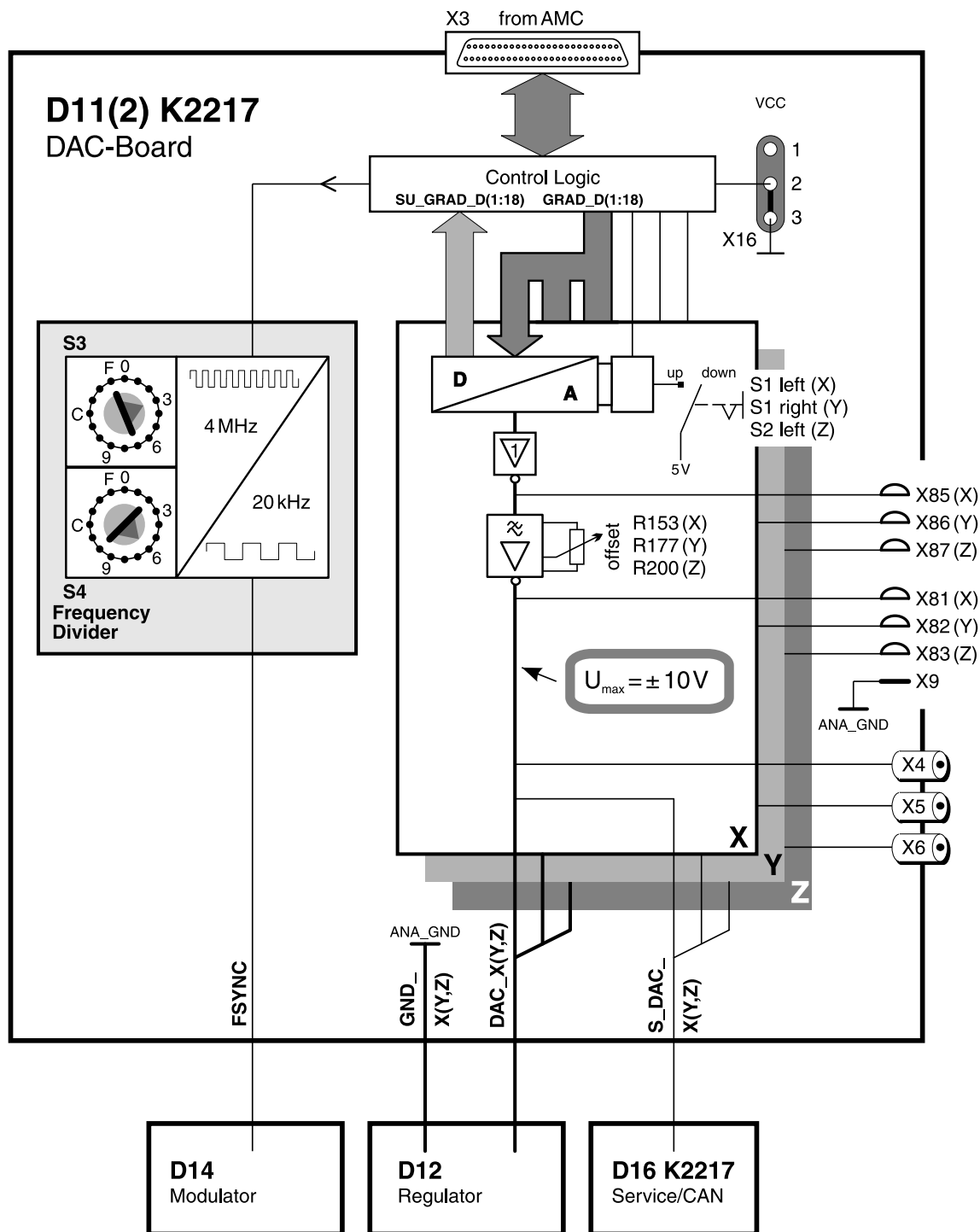
## 2.2 Backplane D18

### 2.2.1 Function

The backplane is the interconnection between the components of the GSSU. It is part of the GSSU rack that is mounted to the cabinet frame by four quick release screws.

## 2.3 DAC Board D11 or D112

Fig. 5: DAC board D11(2)



On newer GPAs, the DAC D11 was replaced by a DAC D112 board (fully compatible). The D11 description is valid for the D112 as well.

### 2.3.1 General

The DAC board can be divided into two functional blocks:

- the DACs for the X, Y, Z channels and
- the frequency divider.

### 2.3.2 Function

The digital data coming from the AMC (Advanced Measurement Control) is converted into analog signals (nominal current value) used to control the regulated current generators. Additionally, the frequency divider implemented on the DAC board generates synchronization signals for components in the GPA.

#### 2.3.2.1 Control Data

The gradient control data coming from the AMC is connected to a 50 pin female front connector (X3). For the detailed description and the timing of the gradient control data, refer to the AMC description.

#### 2.3.2.2 Amplitude data

The 18 bit amplitude data (G\_DAT0 - G\_DAT17) for the three gradient axes is transferred multiplexed and controlled by the select signals G\_SEL\_X, Y and Z. This data contains the eddy current pre-distortion and the gradient delay.

#### 2.3.2.3 Synchronization Clock

Additionally, a 4 MHz clock signal is connected to the board via the same connector. This clock signal is divided by a frequency divider. The division ratio is adjusted by the two rotary switches S3 and S4 resulting in 20 kHz.

#### 2.3.2.4 Loop back Test

The signal Test-Read that is a part of the control data allows a data loopback test. The 18 bit data of the channel X and the control lines can be transferred back to the AMC.

#### 2.3.2.5 Nominal Value

The 18 bit amplitude data of the active channels (X, Y, and Z) is transferred to the three DACs controlled by the select signals. Jumper X16 determines the usage of the gradient data (sign, 2-complement).

The output voltage of the DACs (DAC\_X, Y and Z) is  $\pm 10$  V at full scale. It corresponds to maximum current (depends on jumper setting on regulator D12) driven through the gradient coils.

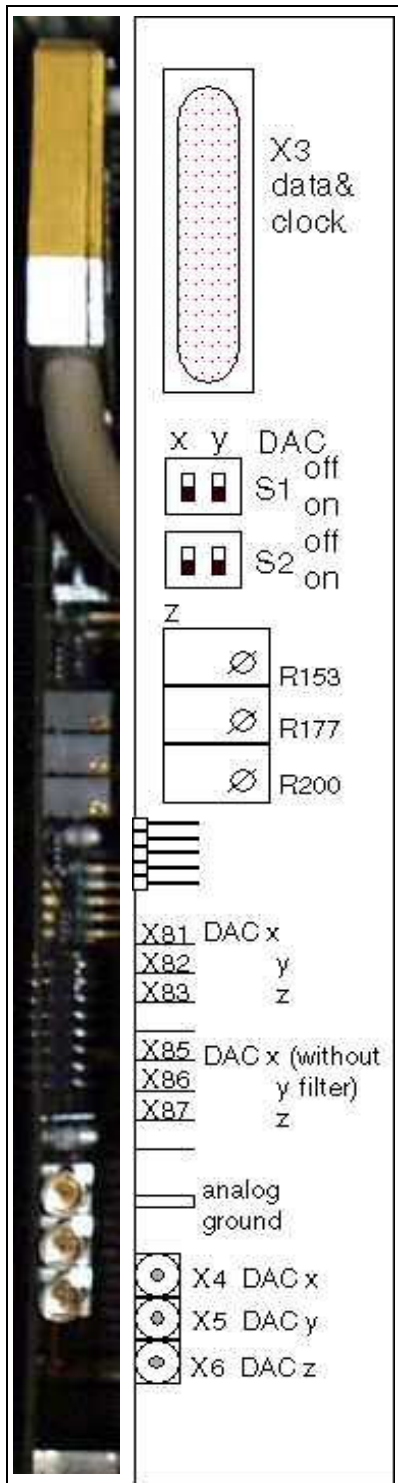
Switches S1 (left/right) and S2 (left) can be used to apply digital zero to the input of the DACs. This is necessary for offset adjustment using potentiometers R153(X), R177(Y), and R200(Z).



### 2.3.2.6 Monitoring

The DAC output signals (DAC\_X, Y, Z) are available for monitoring via QLA-connectors (X4, X5 and X6) and test pins (X81-X83 and X85-X86; GND on X9) on the front panel of the board and they are also connected to the loop multiplexer on the D16 Service/CAN board (S\_DAC\_X, Y, Z).

Fig. 6: Front of DAC-board D11(2)



## Descriptions:

X3:

18 bit parallel data with the Select signals,  
4 MHz synchronization clock and Test-Read signal

S1/S2:

Digital zero to the DAC input  
(S2/right is not implemented)

R153, R177, R200

Potentiometers for offset adjustment

Test pins for measuring signals with oscilloscope

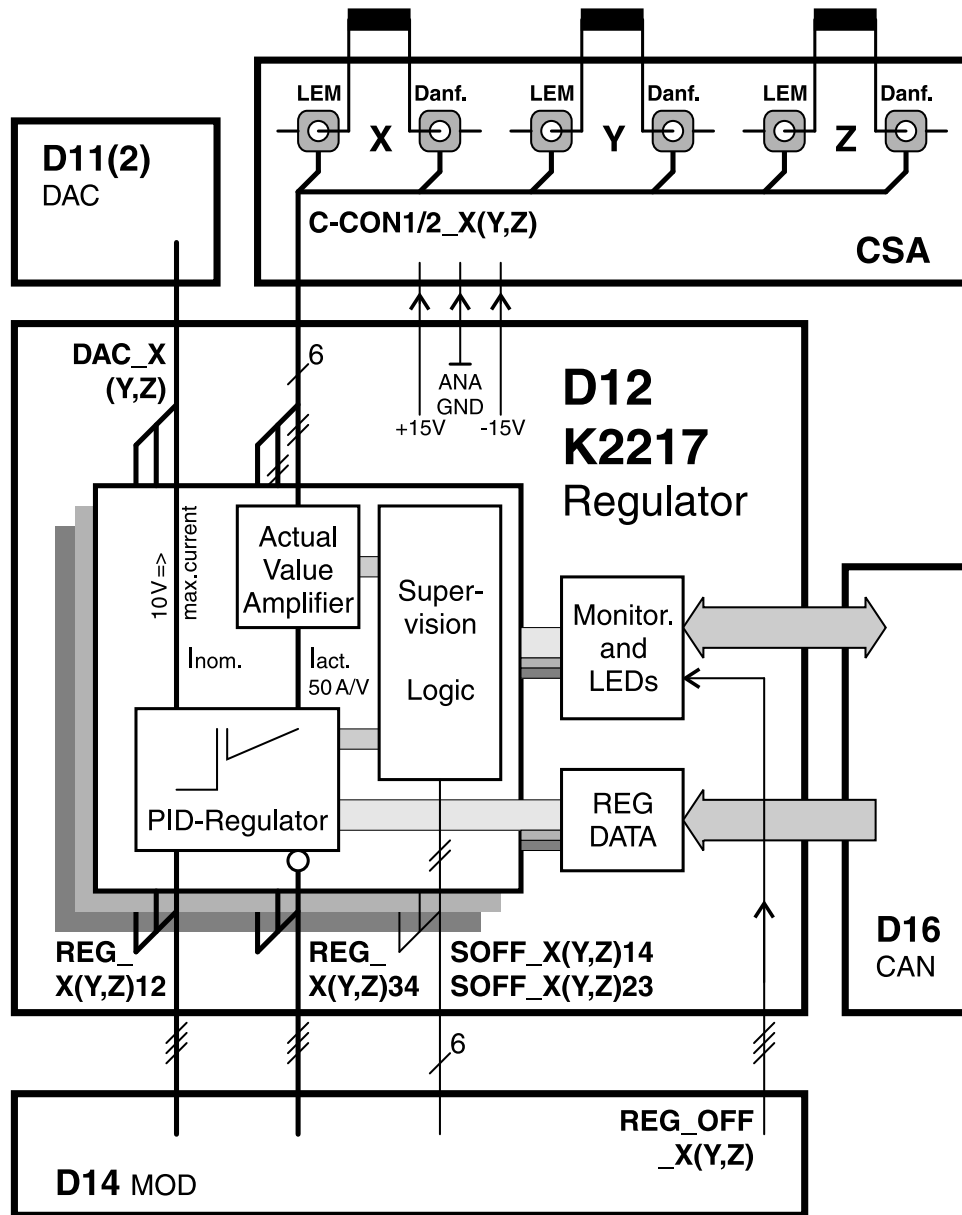
DAC output X4, X5, X6

Nominal value +/-10 V

(10 V corresponds to max. current, depending on the  
setting/jumper on D12)

## 2.4 Regulator Board D12

Fig. 7: Regulator Board D12



### 2.4.1 General

The regulator board includes the regulator function for all three axes. It contains the following functional blocks:

- Actual value amplifier
  - PID regulator
  - Supervision logic
- 
 Regulator unit (3 times)

- Monitoring logic and LED display
- Regulator data logic (REGDATA)

### 2.4.2 Function

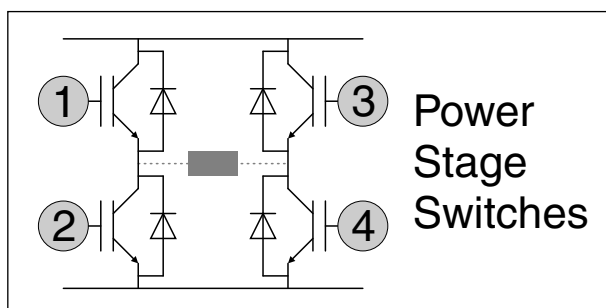
The current sensors in the Current Sensor Assembly (CSA) deliver a compensation current proportional to the current driven through the gradient coil. This compensation current is converted into a corresponding voltage level by the actual value amplifier.

In the PID regulator, the actual current value is compared to the nominal current value  $DAC\_X(Y, Z)$  from the DAC board. The resulting regulator difference is amplified by the regulator according to its characteristic. The output signals  $REG\_X(Y, Z)_{12}$  and  $REG\_X(Y, Z)_{34}$  of the regulator are connected to the input of the modulator board D14.



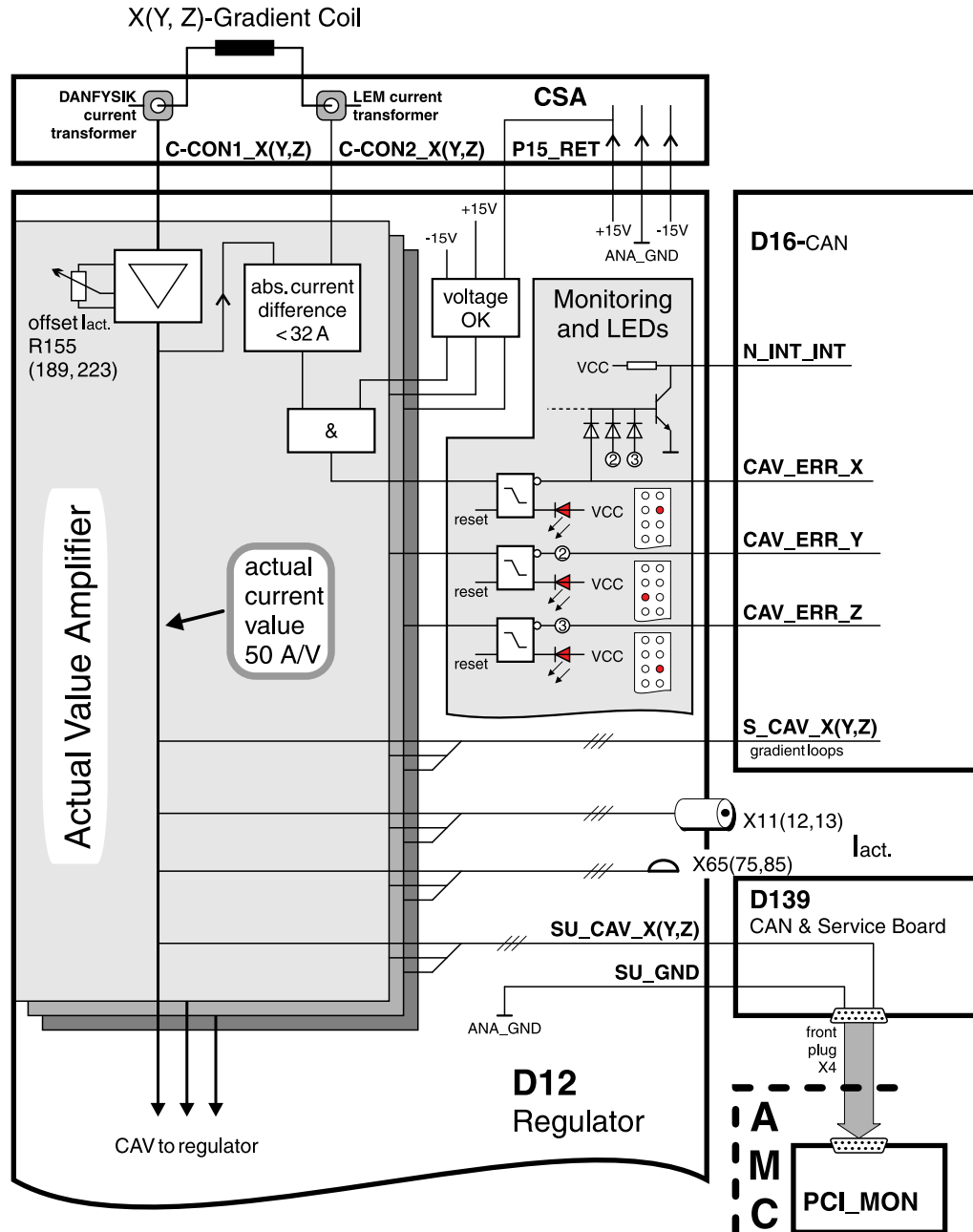
The numbers 12, 34, 14 or 23 in the signal names indicate the power stage switches influenced by the corresponding signal.

Fig. 8:



## 2.4.2.1 Actual Value Amplifier (AVA)

Fig. 9: Actual Value Amplifier



## Amplifier

The measurement current (C-CON1) from the DANFYSIK current transformer is amplified so that 50 A coil current corresponds to 1 V. The signal from the LEM current transformer (C-CON2) is used for monitoring only. The offset of the amplifiers is adjusted by the potentiometers R155(X), R189(Y), and R223(Z). The output signal of the amplifiers (CAV) is available on the front panel for monitoring on QLA test points X11(X), X12(Y), and X13(Z). Test pins X65(75, 85) show the same signal.

### Current Transformer Monitoring

The two signals from the current transformers are checked whether they are the same and do not differ by more than 32 A absolute. The power supply unit ( $\pm 15$  V and ANA\_GND) for the current transformers on the current sensor assembly (CSA) is coming from the D12/124 board and the +15 V is looped back as P15\_RET-signal. This signal is combined with the  $\pm 15$  V voltage monitoring. If one of the above monitoring signals is faulty, the respective current error (CAV\_ERR) is sent to the D16-board. This error signal also lights a red LED on the front panel of D12 and initiates the interrupt internal signal (N\_INT\_INT) that blocks all three axes.

### Loop Test

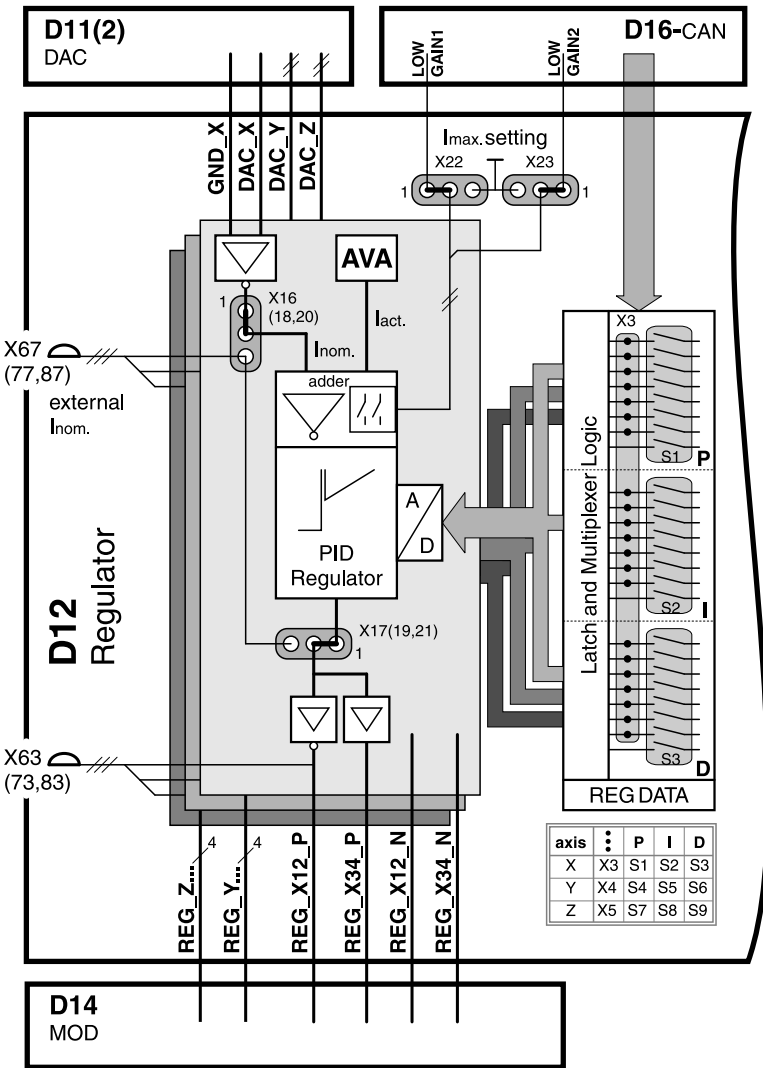
The analog signal S\_CAV is connected to the loop multiplexer on the Service/CAN board for automatic testing.

### Stimulation Monitoring

The actual current value is an important signal for the Software stimulation monitor of the PCI\_MON in the Advanced Measurement Control (AMC). Therefore, the SU\_CAV signal is sent to the D139 CAN & Service Board, where it is digitized and finally fed to the PCI\_MON board of the AMC.

2.4.2.2 PID Regulator

Fig. 10: PID regulator



Current Configuration

The nominal value sent from the DAC board is inverted and via jumper X16(18, 20) added to the actual value coming from the AVA. The two jumpers X22 and X23 are used to set the maximum current value for the gradient system in use:

| X23 | X22 | I <sub>max.</sub>   | Software  |
|-----|-----|---------------------|-----------|
| 1-2 | 1-2 | Software controlled | NUMARIS/4 |

Regulator Unit

The closed loop regulation is based on a digitally controlled PID regulator with 7 bit resolution for the proportional (P), the integral (I), and the differential (D) part.

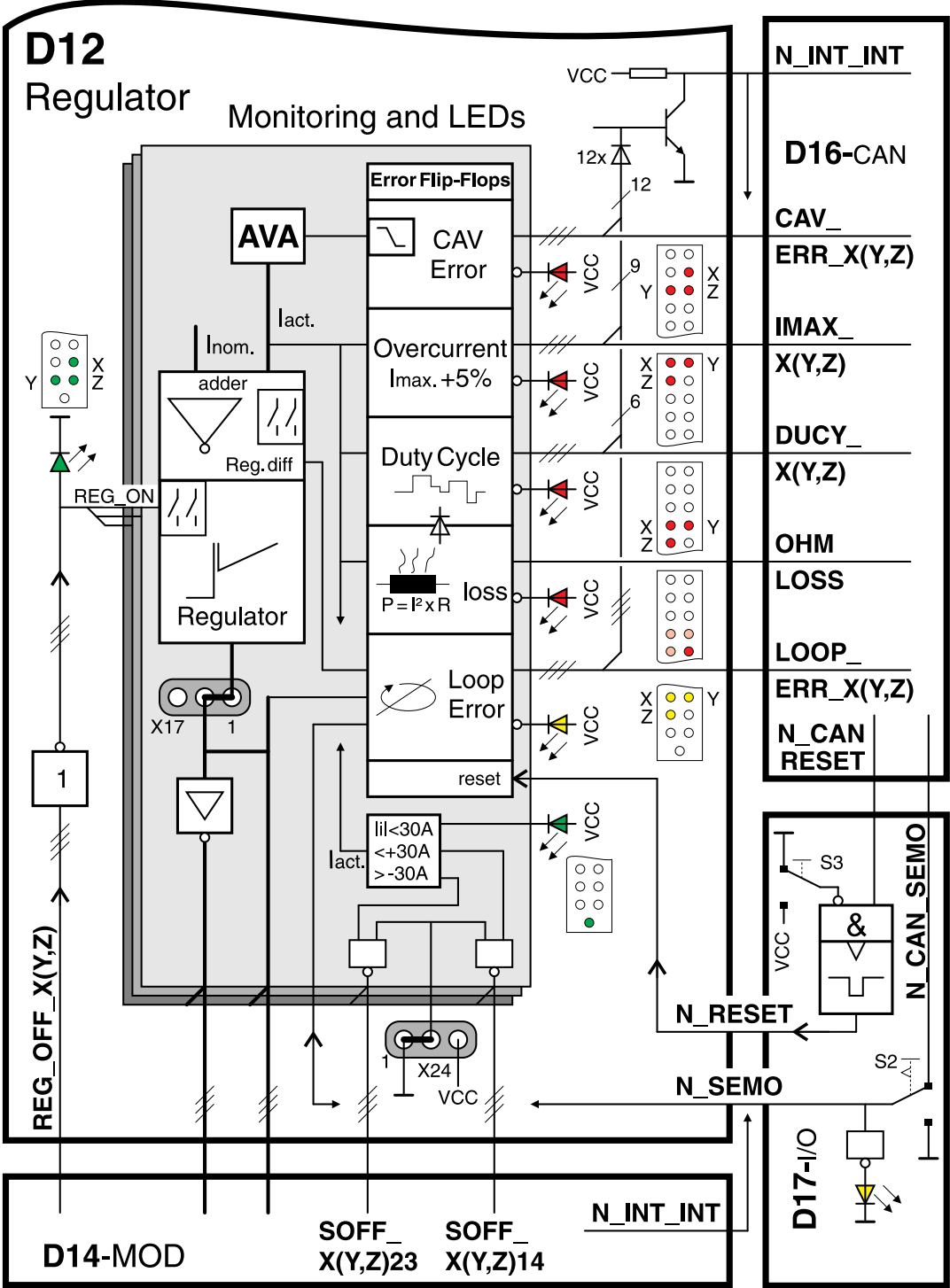
The initial configuration of the regulator after power on is determined by DIP switches S1 to S9 (see table on the left). The nominal configuration, a result of the regulator adjustment procedure, is activated by Software.

The regulator output signal passes jumper X17(19, 21) and is sent to the D14 board (inverted and non-inverted). The inverted regulator output signal can be measured on test point X63(73, 83). Test point X67(77, 87) can be used to supply an external nominal value (via X16) or an external regulator signal (via X17).



2.4.2.3 Monitoring

Fig. 11: Monitoring on Regulator Board



Errors and LEDs. The following table shows the monitoring on the D12 board:

Tab. 2 Error monitoring and LED display

| Error       | Signal monitored                                           | LEDs     | INT_INT                     |
|-------------|------------------------------------------------------------|----------|-----------------------------|
| CAV error   | Current sensors<br>15 V return from CSA; $\pm 15$ V supply | 3 red    | Yes                         |
| Overcurrent | Actual current value                                       | 3 red    | Yes                         |
| Duty cycle  | Actual current value                                       | 3 red    | Yes                         |
| Ohm loss    | Actual current value                                       | 1 red    | No                          |
| Loop error  | Regulator difference<br>Regulator output signal            | 3 yellow | Yes, if not in service mode |

The interrupt internal (N\_INT\_INT) always blocks all three axes for safety reasons.

An ohm loss error additionally results in a duty cycle error for identification of the faulty axis.

### Reset

All of the above mentioned errors are latched in error flip-flops and sent to the D16 board. The error flip-flops are reset by Software via the D16 board or manually using button S3 on the D17 board.

### Service Mode

The service mode can be activated either by Software via the D16 board or by switch S2 on the D17 board. The service mode is indicated by a yellow LED on the D17 board. If service mode is activated, the loop error does not result in an internal interrupt. Nevertheless, the yellow LEDs will light and the error is forwarded to the D16 board.

### Regulator enable

The modulator board D14 sends the signal REG\_OFF to enable and disable the regulator board. The enable status is shown by three green LEDs on the front panel of the regulator board.

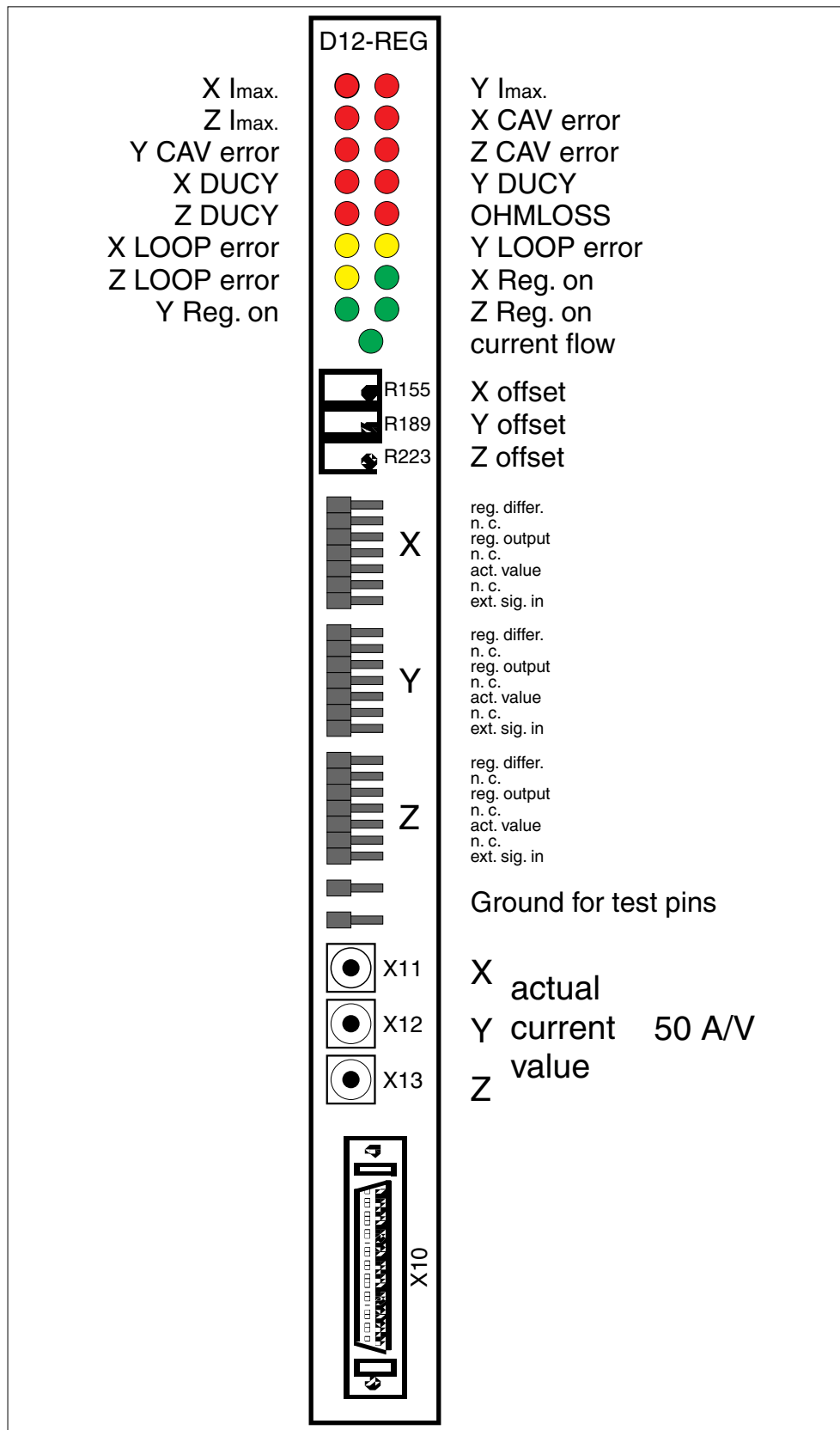
### Opposite Current Direction disabled

The actual value is monitored for a threshold of approximate 30 A absolute. A green LED indicates if a current of more than +30 A or less than -30 A is running. Furthermore, the other two outputs of this comparator logic ( $< 30$  A and  $> -30$  A) are used to disable the transistors of the opposite current direction.

The signal SOFF\_X(Y, Z)23 disables the positive current direction and the signal SOFF\_X(Y, Z)14 disables the negative current direction. Both signals are sent to the modulator board D14.

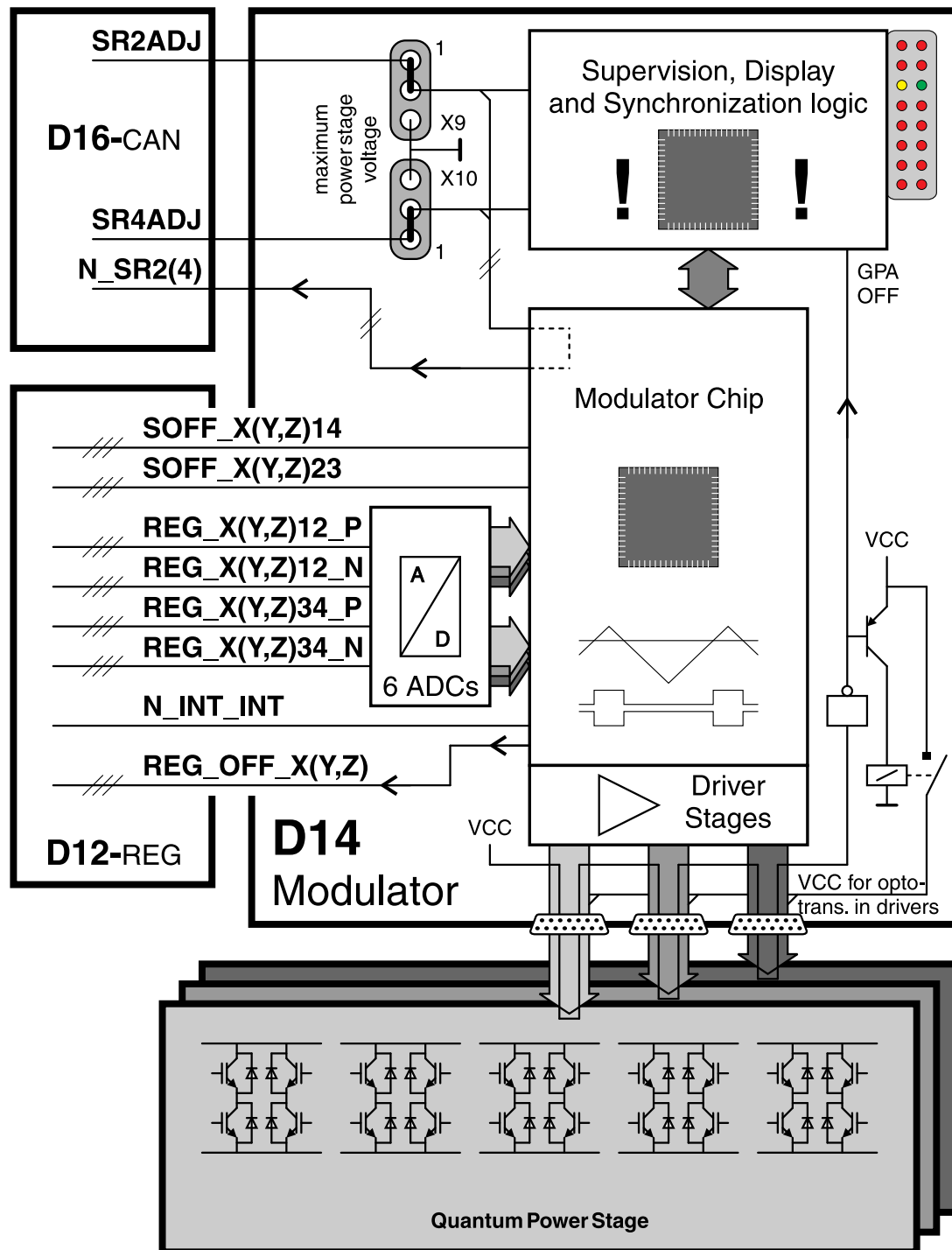
Jumper X24 determines whether this disable logic for the opposite current direction is used (1-2) or not (2-3).

Fig. 12: Front Panel of Regulator Board D12



## 2.5 Modulator Board D14

Fig. 13: Modulator Board D14



### 2.5.1 General

Two main functions are implemented on the Modulator Board D14:

- the generation of the pulsewidth modulated switching signals for the three axes;

- the supervision of power stage related signals for the three power stages.

## 2.5.2 Function

### 2.5.2.1 Pulsewidth Modulator

The D14 board is based on a programmable logic that contains the pulsewidth modulator and the monitoring circuit of time critical to power-stage related signals.

The regulator output signals REG\_X(Y, Z)\_12(34)\_P(N) from the regulator board D12 are analog-to-digital converted and fed to the modulator chip where the pulsewidth modulation takes place. The switch signals (low-active) for the power stage transistors are fed out by a front plug for each of the three axes. By means of a standard SCSI cable (it is NOT a SCSI bus!), the signals are connected to the power stage modules. The connection of those SCSI cables is monitored by a bridge between pin 1 and pin 50 inside the SCSI plug in the power stage modules. A 5 V signal (VCC) from the modulator board is routed via the three plugs and monitored by the supervision logic (GPA\_ON). If this signal is missing, the VCC supply for the opto-transmitters on the driver boards in the power stages is switched off by means of a relay.

The signals SOFF\_X (Y, Z)14(23) inform the modulator chip whether the opposite current direction should be enabled or disabled. The N\_INT\_INT signal from the regulator board informs the modulator chip about a severe problem of the regulator. Consequently, the modulator disables the regulator of the desired axis by means of signal REG\_OFF\_X(Y, Z).

### 2.5.2.2 Maximum Power Stage Voltage

The max. power stage voltage is selected by Software, therefore jumper X9 and X10 must be set up accordingly:

Tab. 3 Power stage voltage setting by jumper X9 and X10

| X9              | X10             | Max. power stage voltage | Software  |
|-----------------|-----------------|--------------------------|-----------|
| 1–2<br>(SR2ADJ) | 1–2<br>(SR4ADJ) | Software controlled.     | NUMARIS/4 |

The actual configuration of the power stage voltage is fed back to the processor board D16 CAN by the modulator chip. If the power stage voltage is set to >1000 V, the gradient system requires a stimulation monitor. Systems running with NUMARIS/4 use a Software stimulation monitor.

### 2.5.2.3 Current and Voltage Combinations

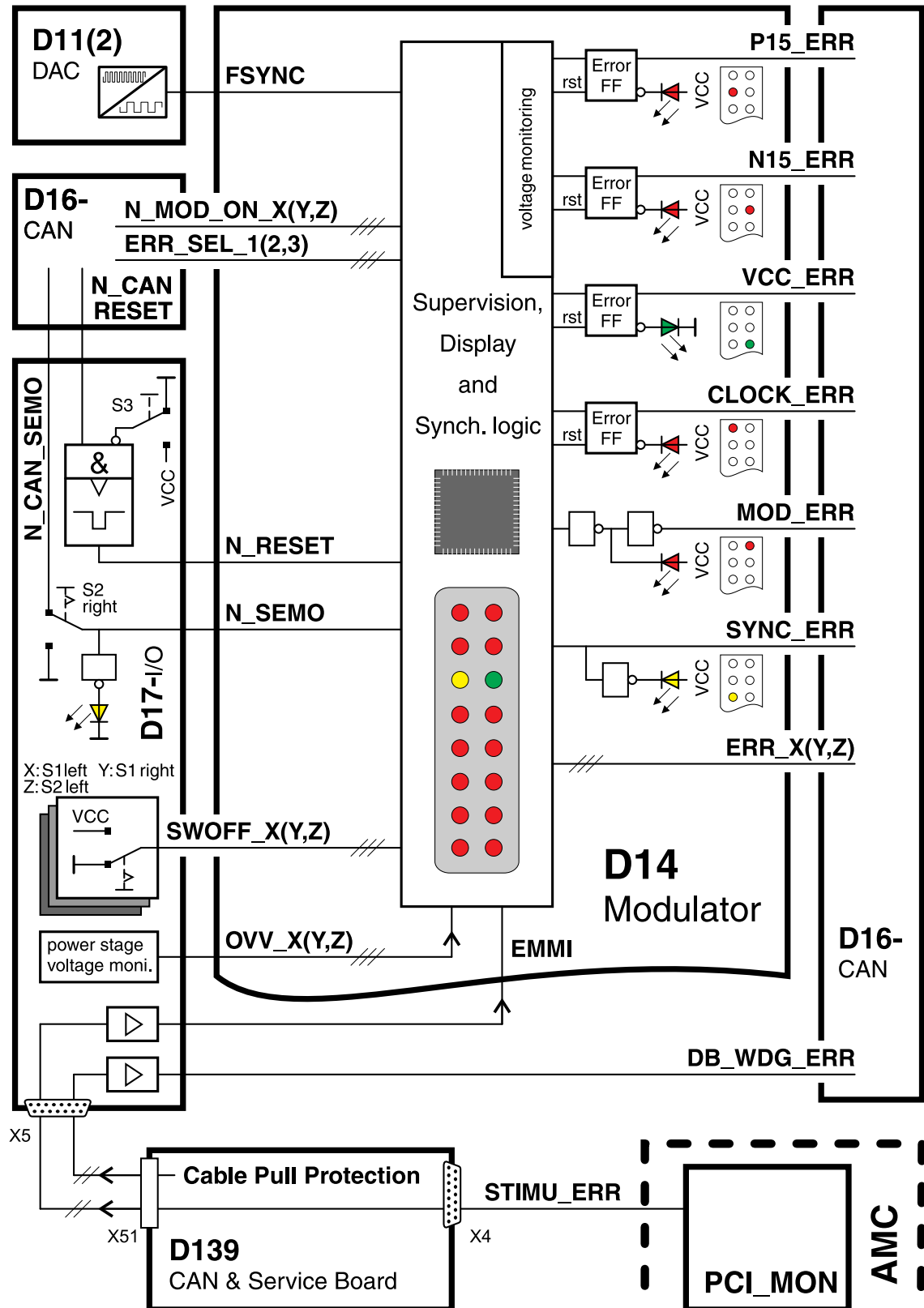
The following current and voltage combinations are used at present:

Tab. 4 Current and voltage combinations

| Current | Voltage | MAGNETOM system      |
|---------|---------|----------------------|
| 380 A   | 2000 V  | Symphony Tim Upgrade |

### 2.5.2.4 Supervision, Display and, Synchronization Logic

Fig. 14: Supervision on D14



### Monitoring

The supervision monitors the supply voltages ( $\pm 15\text{ V}$  and  $+5\text{ V} = \text{VCC}$ ), clock, and synchronization. Furthermore, monitoring signals from other boards can disable the modulator:

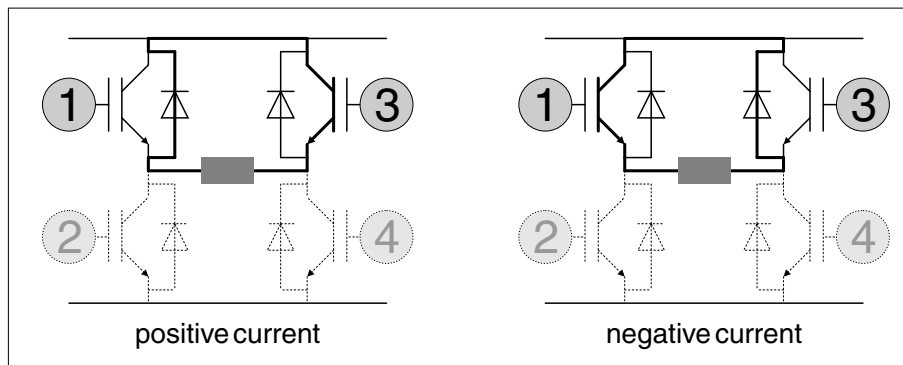
- Software Stimulation Monitor from AMC
- Power stage voltage monitoring from D17
- Manual modulator off by switches S1 and S2 on D17
- Modulator enable/disable from the processor board D16.

If the modulator detects an error, the processor board D16 is informed via signal  $\text{ERR\_X(Y, Z)}$  and consequently, asks the modulator supervision for the reason using signal  $\text{ERR\_SEL\_1(2, 3)}$ .

### Switch-off Modes

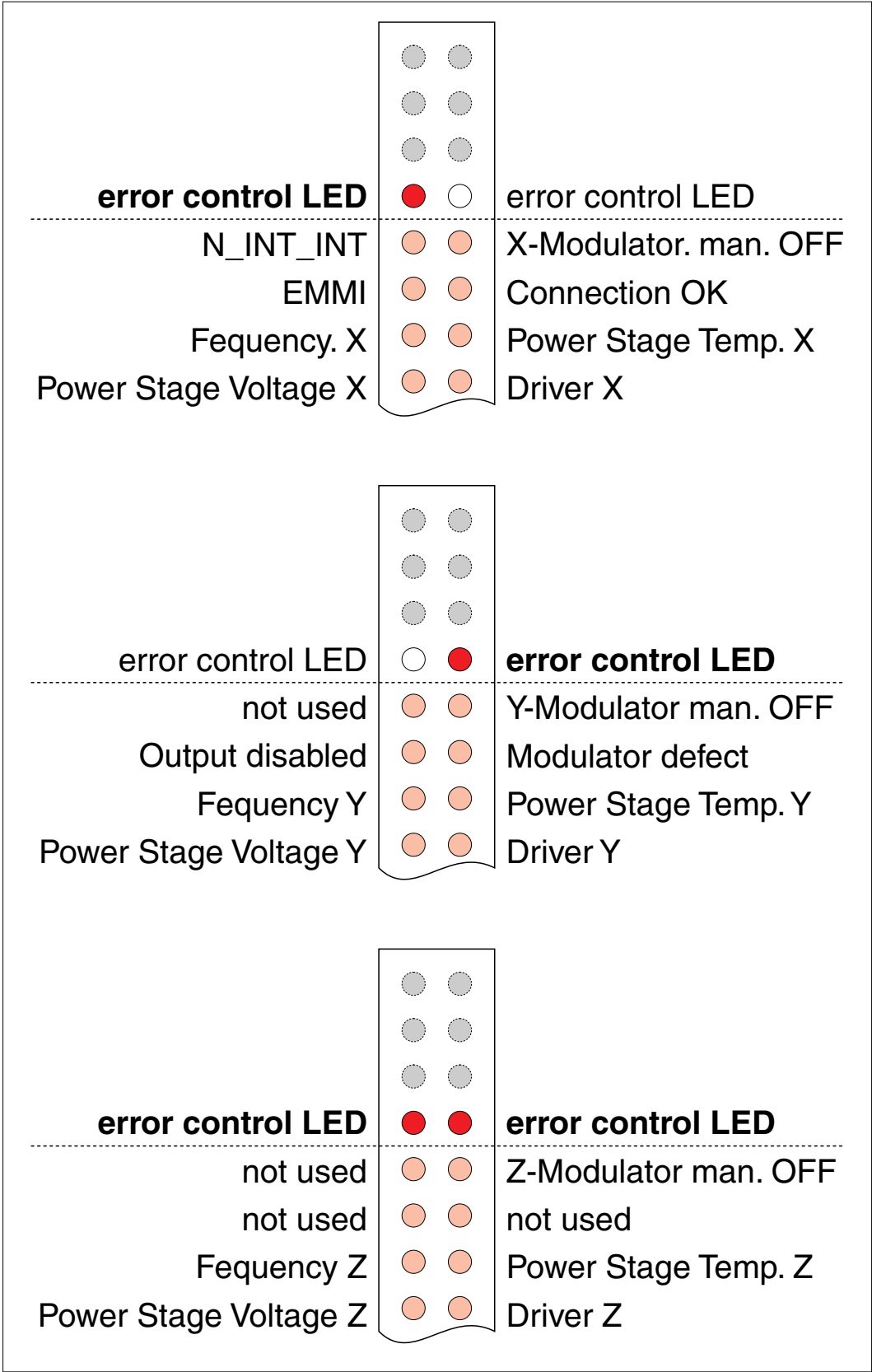
In case of an error, the modulator can disable a running gradient current in two ways:

- The "hard" way; all switches of the power stage are blocked.
- The "soft" way; the modulator enables switch 1 and 3 of the power stage permanently (upper freewheeling).



This "soft off" is necessary to prevent stimulation during switch off. Only the processor-controlled modulator disables using signal  $\text{N\_MOD\_ON\_X(Y, Z)}$ . The manual off (SWOFF...) by the switches on the D17 board result in a hard switch off of the gradient current.

Fig. 15: Error Display D14





### LED Display

The different errors are indicated by eight error LEDs and two control LEDs. The status of the two control LEDs (LED 7 and 8) determines the error indicated by the lower eight LEDs. (see drawing on the left)

### Reset of errors

All errors can be reset by the processor D16 (N\_CAN\_RESET) or manually by switch S3 on I/O board D17.

### Service Mode

The supervision logic can be switched into service mode (N\_SEMO) by front panel switch S2\_right on the D17 board and remotely from the D16 board (N\_CAN\_RESET). The service mode is indicated by a yellow LED on the front panel of the I/O board D17.

## 2.6 Service/CAN Board D16 K2217

Fig. 16: Service/CAN board D16

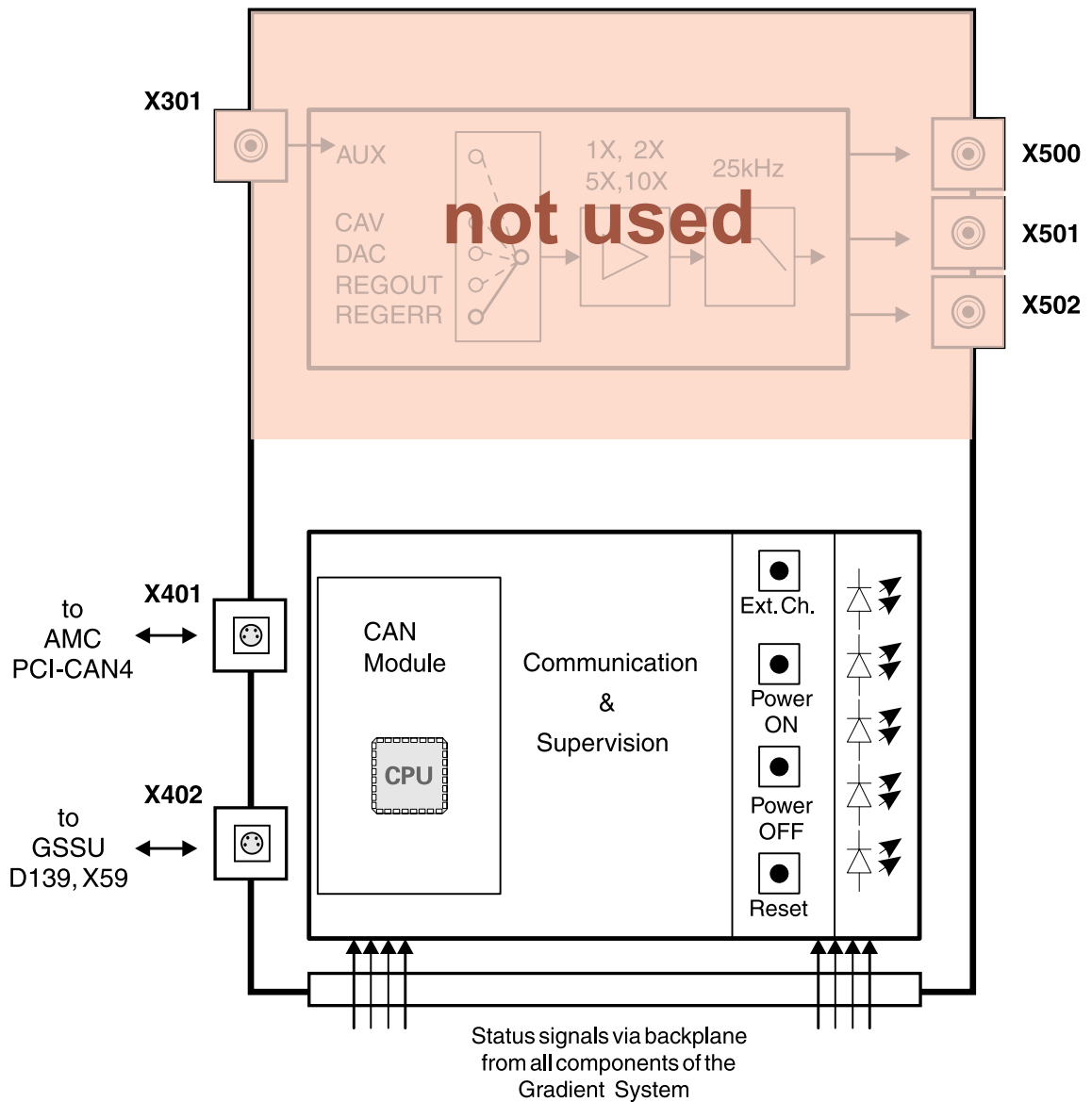
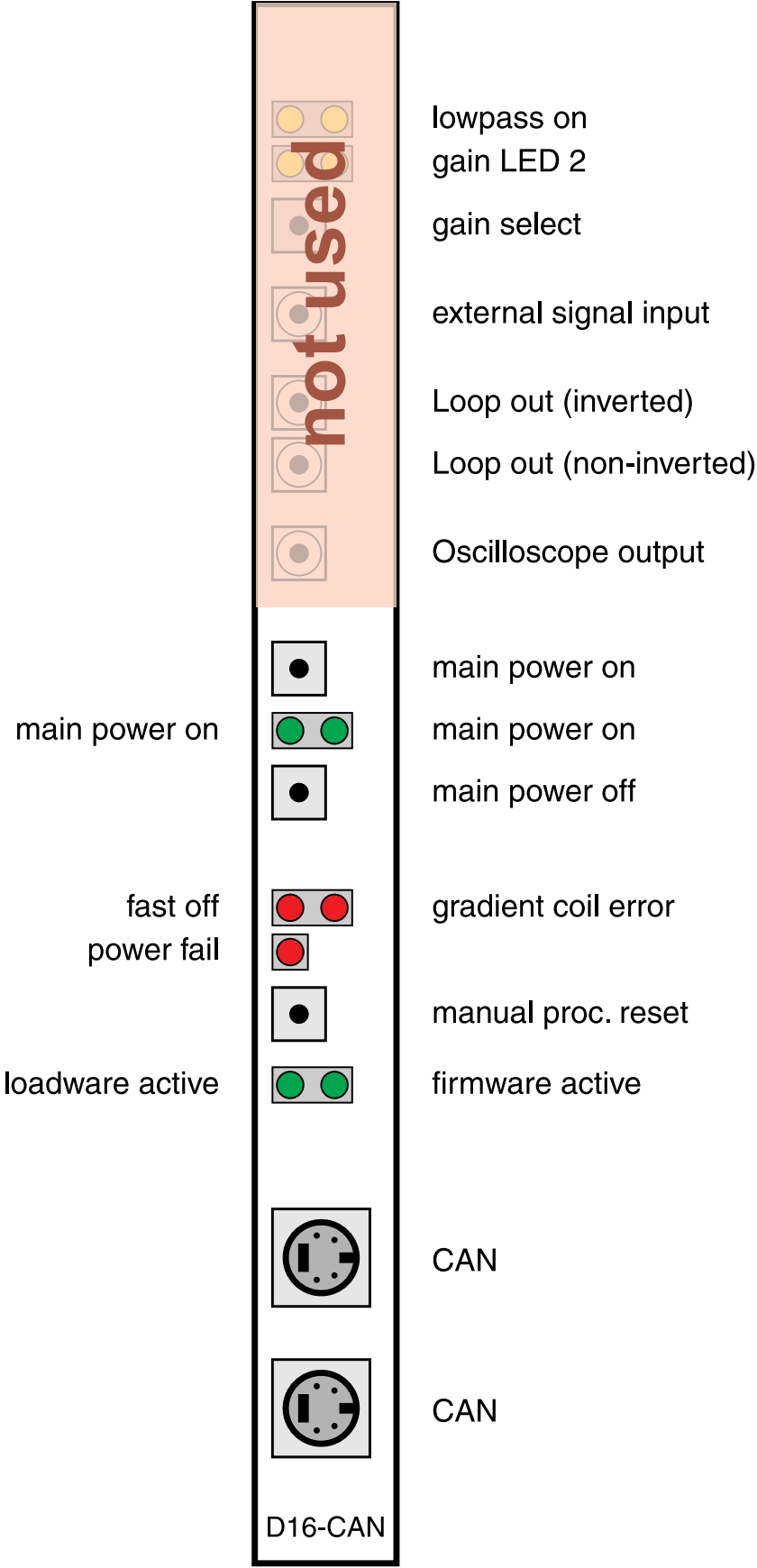


Fig. 17: Front panel of Service/CAN board D16



### 2.6.1 Function

The D16 Service/CAN board consists of two functional blocks:

- **CAN module** - Responsible for communication and supervision
- **Signal multiplexer (not used on Symphony Tim Upgrades)** - Instead of this analog multiplexer, the D139 board supplies a digital signal loop for gradient signals to the AMC.

#### 2.6.1.1 CAN Module

##### Communication

The CAN module is responsible for data transmission between the AMC and the GPA. Control data for the GPA is transferred from the AMC and GPA status information back to the AMC.

The firmware (FW) stored in a PROM allows the processor to run a self-test and to communicate and load the application software over the CAN bus. The application software - i.e. loadware (LW) - for the processor is stored in an EEPROM.

### 2.6.2 Switches

Tab. 5 D16 switches

| Switch            | Description                            |
|-------------------|----------------------------------------|
| S401 (pushbutton) | Not used                               |
| S402 (pushbutton) | Switches on the GPA power supply unit  |
| S403 (pushbutton) | Switches off the GPA power supply unit |
| S404 (pushbutton) | Processor reset                        |

### 2.6.3 LEDs

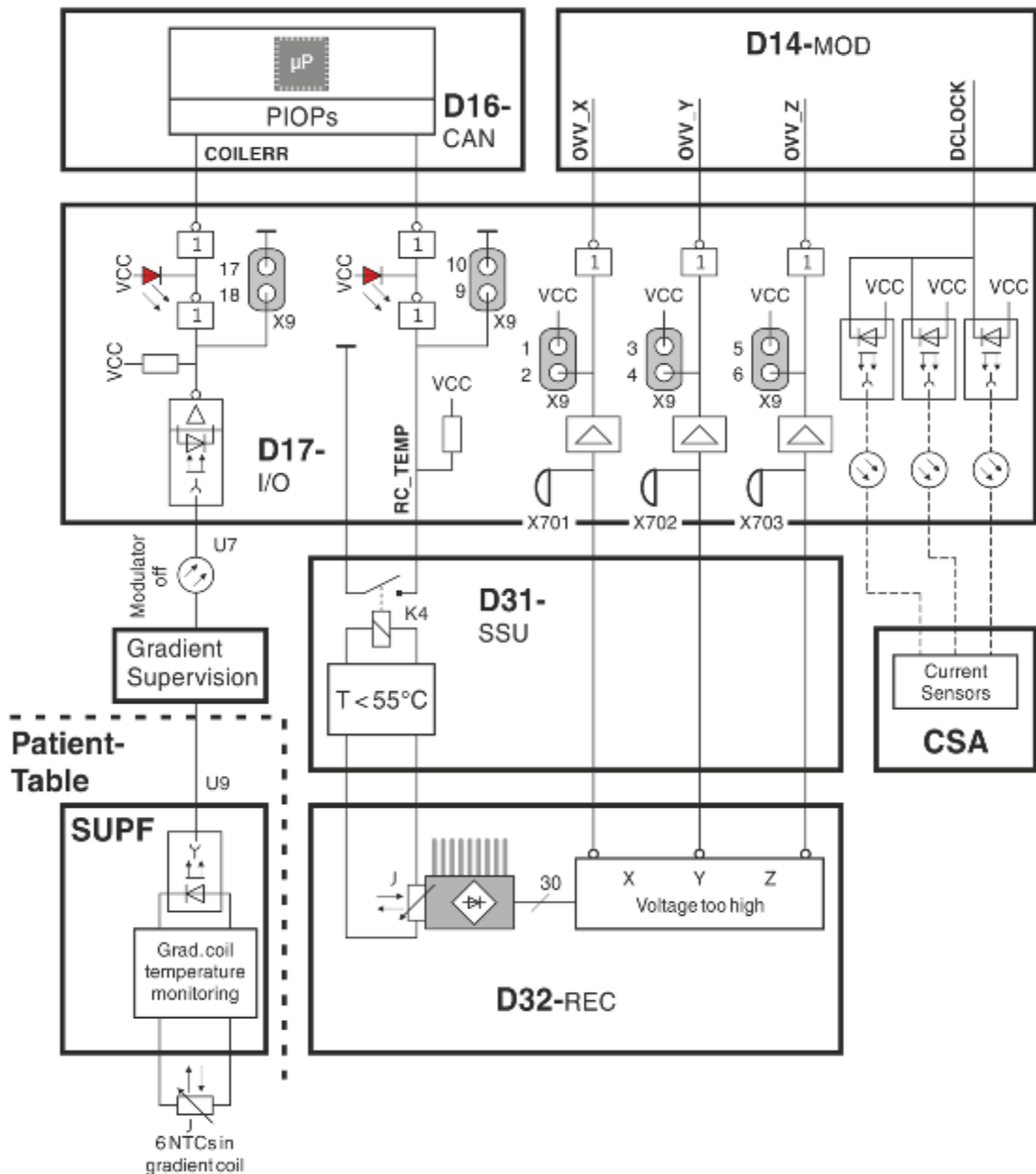
Tab. 6 D16 LEDs

| LEDs<br>(from top to bottom) | Description                           |
|------------------------------|---------------------------------------|
| V408, V412 (yellow)          | Not used                              |
| V407 (green)                 | Main power on                         |
| V413 left (red)              | GPA fast off (emergency off)          |
| V413 right (red)             | Gradient coil error (overtemperature) |
| V607 (red)                   | Power fail                            |

| LEDs<br>(from top to bottom) | Description     |
|------------------------------|-----------------|
| V403 left (green)            | Loadware active |
| V403 right (green)           | Firmware active |

## 2.7 I/O Board D17 K2217

Fig. 18: I/O board D17



### 2.7.1 General

The D17 I/O board is an interface between the GSSU and the power stage section. Control and monitoring signals are fed via this board to the modulator board D14 or the Service/CAN board D16.

## 2.7.2 Function

### 2.7.2.1 Power Stage Voltage Monitoring

Each power stage module needs 5 times 400 VDC for the five cascades. Rectifier board D32 holds 15 rectifier units for the three axes. The over voltage is monitored. The three error signals pass the soft start unit D31 and block the modulator via the I/O board.

### 2.7.2.2 Gradient Temperature Monitoring

The following temperature-related signals are monitored via the D17 I/O board:

- **Rectifier Temperature** - the 15 rectifier units on the D32 board are mounted on a water-cooled heat sink, the temperature of which is monitored by a PTC resistor. A monitoring electronic on D31 checks if the temperature of the rectifier heat sink is below 55°C and activates relay K4. In case of an error, relay K4 is deactivated and D17 sends an error signal to the D16 board. A red LED at the D17 indicates this error.
- **Gradient Supervision Alarm** - in case the Gradient Supervision has detected smoke particles (e.g. from overheated gradient connections), the light at FOC D17/U7 will be removed and the GPA will be disabled with Modulator off. In addition, a relay at the VLM opens and via the AMC a warning pop-up is posted to the console (Scanner Hardware Error. Automatic Shutdown! Evacuate patient...).

One minute later, the VLM opens a second relay contact and powers-off the MR-system to stand-by.

- **Gradient Coil Temperature** - the temperature of the Gradient Coil is measured with six thermal resistors (NTCs), moulded inside the Gradient Coil and connected via X50 to the monitoring circuit SUPF (Supplied Functions, A4120-board) at the Patient Table Control Electronics frame. From there, the signal is routed via fiber optic connector U9 to the Gradient Supervision and forwarded to FOC D17/U7.

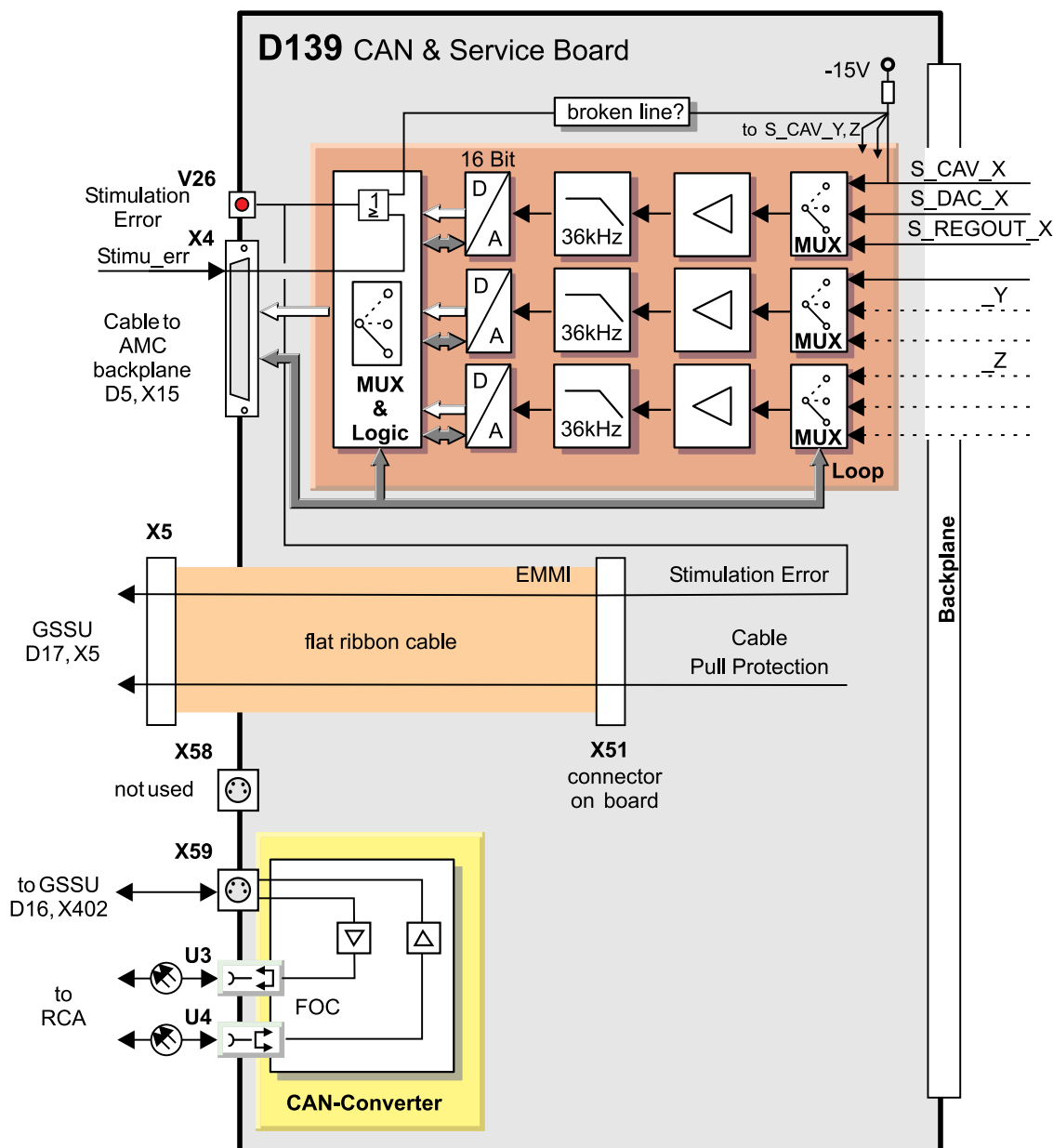
In case of gradient coil over temperature, the light at FOC D17/U7 will be also removed and the GPA will be disabled with modulator off. The GPA will inform the AMC and a warning pops-up at the console (Gradient Power Amplifier Error: "Temperature of the gradient coil is too high").

### 2.7.2.3 Synchronization for DANFYSIK Unit

The signal DCLOCK from the modulator is sent to the DANFYSIK current sensors in the Current Sensor Assembly (CSA) to synchronize these units to the system clock.

## 2.8 CAN & Service Board D139

Fig. 19: D139 CAN & Service Board



### 2.8.1 General

The CAN & Service Board D139 adapts the K2217 GPA to the new architecture of the AMC, supplying the following functions:

- **CAN bus converter** - electrical CAN to optical CAN to the RCA cooler cabinet
- **Digital signal multiplexer (Loop)** - A/D converts GPA signals and supplies them to the AMC



The signal multiplexer is used to connect signals generated in the GPA to the AMC (PCI\_MON board). This link allows Software evaluation of the selected signals for stimulation monitoring and Tune-up as well as tests.

## 2.8.2 Function

### 2.8.2.1 CAN Bus Converter

Due to the new CAN bus layout on Symphony Tim Update systems, an additional converter from electrical CAN to fiber optic CAN is required to supply the RCA cooler cabinet. The electrical CAN is taken from connector X402 of the D16 Service/CAN board and fed to X59 of the CAN & Service Board D139. On-board, the differential signal is converted into two fiber optic CAN signals to the RCA, supplied by the FOC connectors U3 and U4 at the front side of the board.

### 2.8.3 Digital Signal Multiplexer (Loop)

On Symphony Tim Upgrade systems, the analog loop-multiplexer of the D16 Service/CAN board cannot be used any longer due to the different system architecture that requires a digital signal loop to the AMC. The D139 signal multiplexer can select 3 analog GSSU signals from each gradient axis via the backplane:

- **Current actual value** S\_CAV\_X(Y, Z)
- **Nominal value** S\_DAC\_X(Y, Z)
- **Regulator output** S\_REGOUT\_X(Y, Z)

#### 2.8.3.1 Tests and Tune-up

These signals are converted into digital data and fed to the AMC for Software evaluation, e.g. for tests and Tune-up under SeSo.

#### 2.8.3.2 Stimulation Monitoring

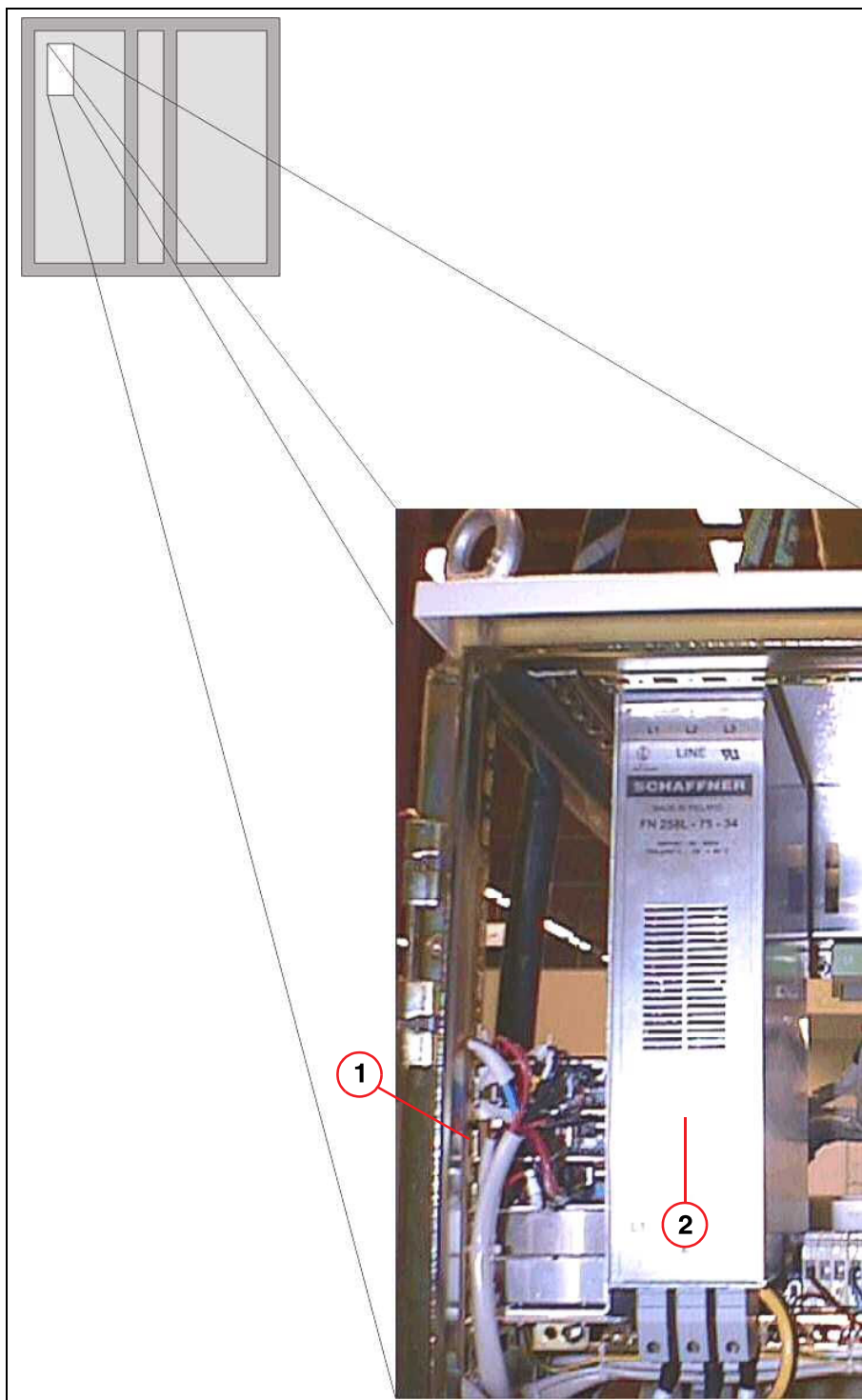
During imaging sequences, the 3 current actual values S\_CAV\_X(Y, Z) are converted into digital data and fed multiplexed via X4 to the PCI\_MON board of the AMC for stimulation monitoring.

In case critical thresholds are exceeded, the PCI\_MON sets the Stimu\_err signal to the D139 and the red "Stimulation Error" LED V26 lights up. The signal is fed via a flat ribbon cable to connector X5 of the D17 I/O-board, which will "soft stop" the modulators immediately to prevent peripheral nerve stimulation of the patient.

For safety reasons, the proper connection of the flat ribbon cable from D139 to D17 is monitored by the "Cable Pull Protection" signal.

### 3.1 Overview

Fig. 20: GSSU Power Supply



- (1) GSSU Power Supplies
- (2) Line Filter

## 3.2 DC Supply for GSSU

### 3.2.1 Function

Stabilized DC voltages for the GSSU. Voltages and tolerances with disabled modulator:

| Voltage   | Tolerance   | Measurement point | Current |
|-----------|-------------|-------------------|---------|
| 5.075 V   | $\pm 25$ mV | D17 X63           | 3 A     |
| 15.050 V  | $\pm 25$ mV | D17 X61           | 2 A     |
| -15.050 V | $\pm 25$ mV | D17 X65           | 2 A     |

### 3.3 DC Supply for the Driver Supply

#### 3.3.1 Function

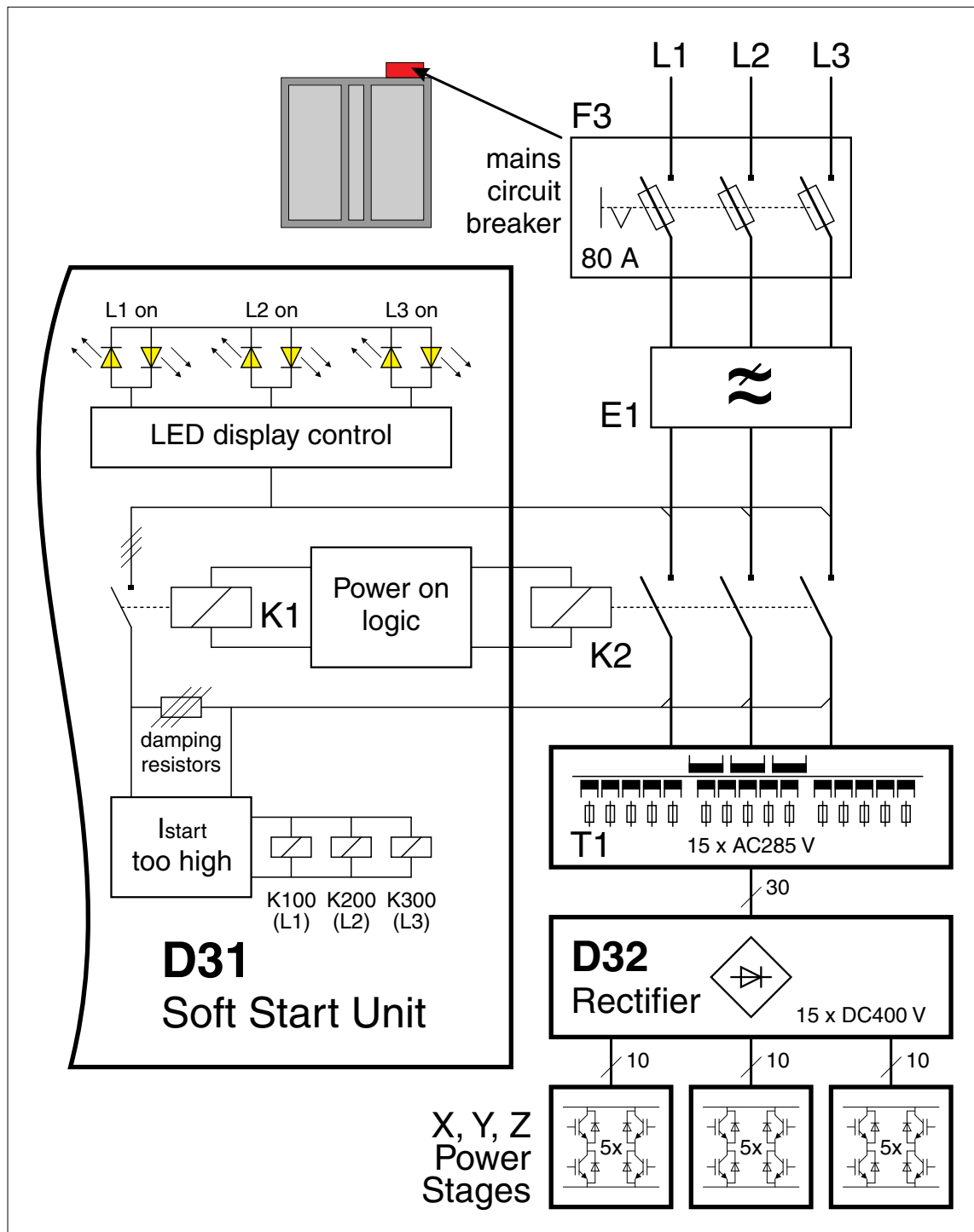
Stabilized DC voltage for the driver supply in the power stage modules.

| Voltage  | Tolerance    | Measurement point          | Current |
|----------|--------------|----------------------------|---------|
| 16.500 V | $\pm 100$ mV | Direct on the power supply | 12 A    |



The stabilized DC voltages are adjustable by potentiometers on the respective power supply.

Fig. 21: Power Stage Supply



## 4.1 Overview

### 4.1.1 General

#### 4.1.1.1 Mains Supply

The mains (L1, L2, L3) are supplied by a 80 A circuit breaker and a line filter. The presence of the mains supply is shown by three yellow LED pairs on the Soft Start Unit D31.

Via contactor K2, this supply is switched to transformer T1, which generates on its secondary output 15 times AC 285 V for the power stage supply. This AC voltage is rectified on the D32 board and distributed to the three power stage modules (5 times 400 VDC each).



To completely remove the power to the gradient system you **MUST** switch off the circuit breaker at the right on top of the cabinet!



The above mentioned circuit breaker **MUST** be switched on first before powering up the gradient system. Otherwise, severe damage to the gradient system is possible!

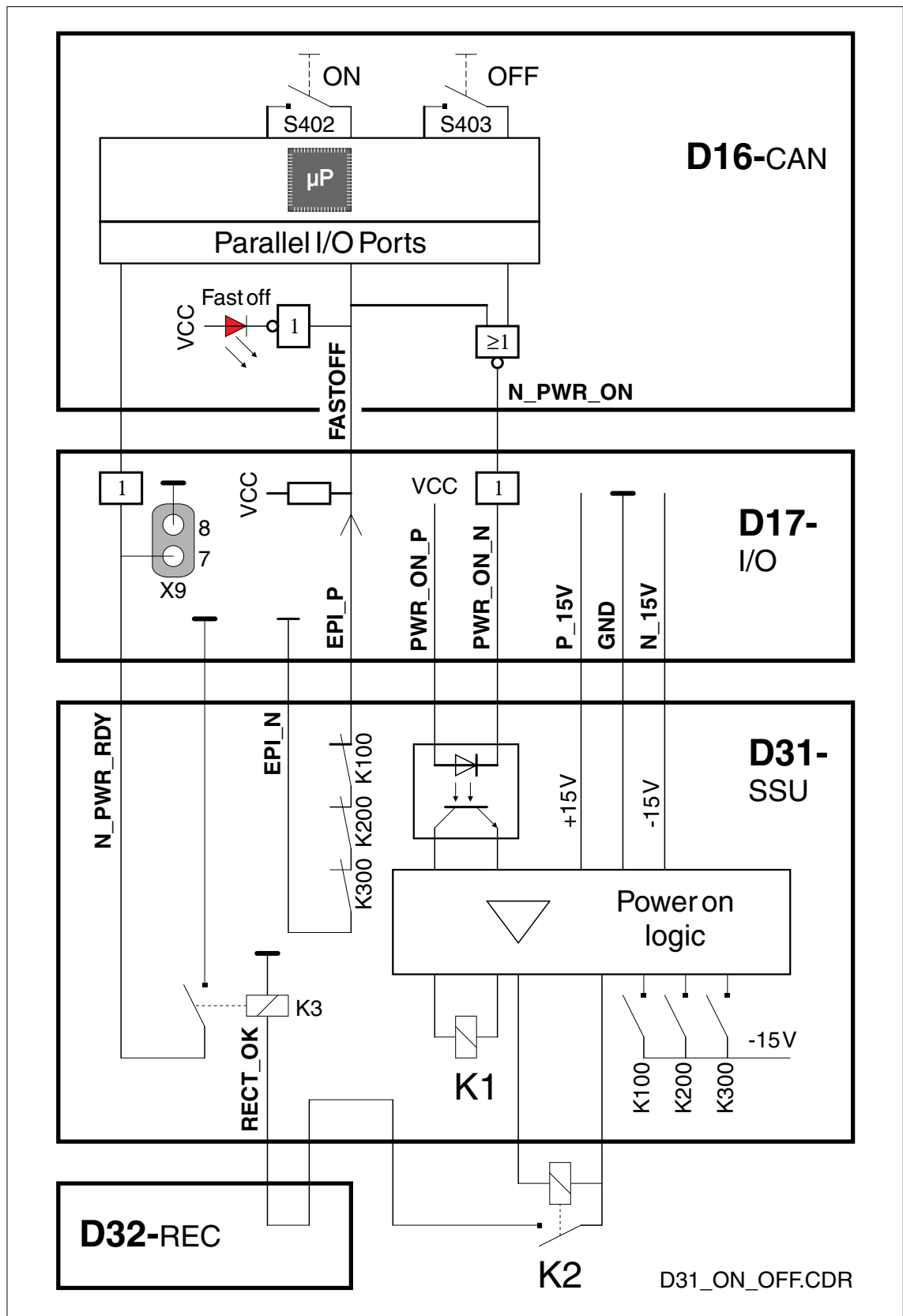
#### 4.1.1.2 Start-up of Power Stage Supply

The power-on logic on D31 energizes the contactor K1 that feeds the mains supply via damping resistors to the transformer. The current via these damping resistors is monitored. If this start-up current is too high, the respective relay (K100 for L1, K200/L2 and K300/L3) is activated and results in a FASTOFF of the power stage supply.

About 2 secs. after K1 is on and no overcurrent is monitored, the power-on logic starts contactor K2 that bridges the damping resistors. The mains supply is now directly connected to the transformer T1.

## 4.2 Soft Start Unit D31 K2217

Fig. 22: Soft Start Unit D31



### 4.2.1 Function

#### 4.2.1.1 Power ON/OFF of GPA

The power ON/OFF command for the gradient power stage supply can either come via software from NUMARIS or by using switches S402 and S403 on the Service/CAN D16.

The power-on signal PWR\_ON\_N (D17) switches on contactor K1 on the D31. In case of an overcurrent in at least one of the damping resistors, the contactor K1 is switched off (K2 not activated) by relay K100, K200 or K300.

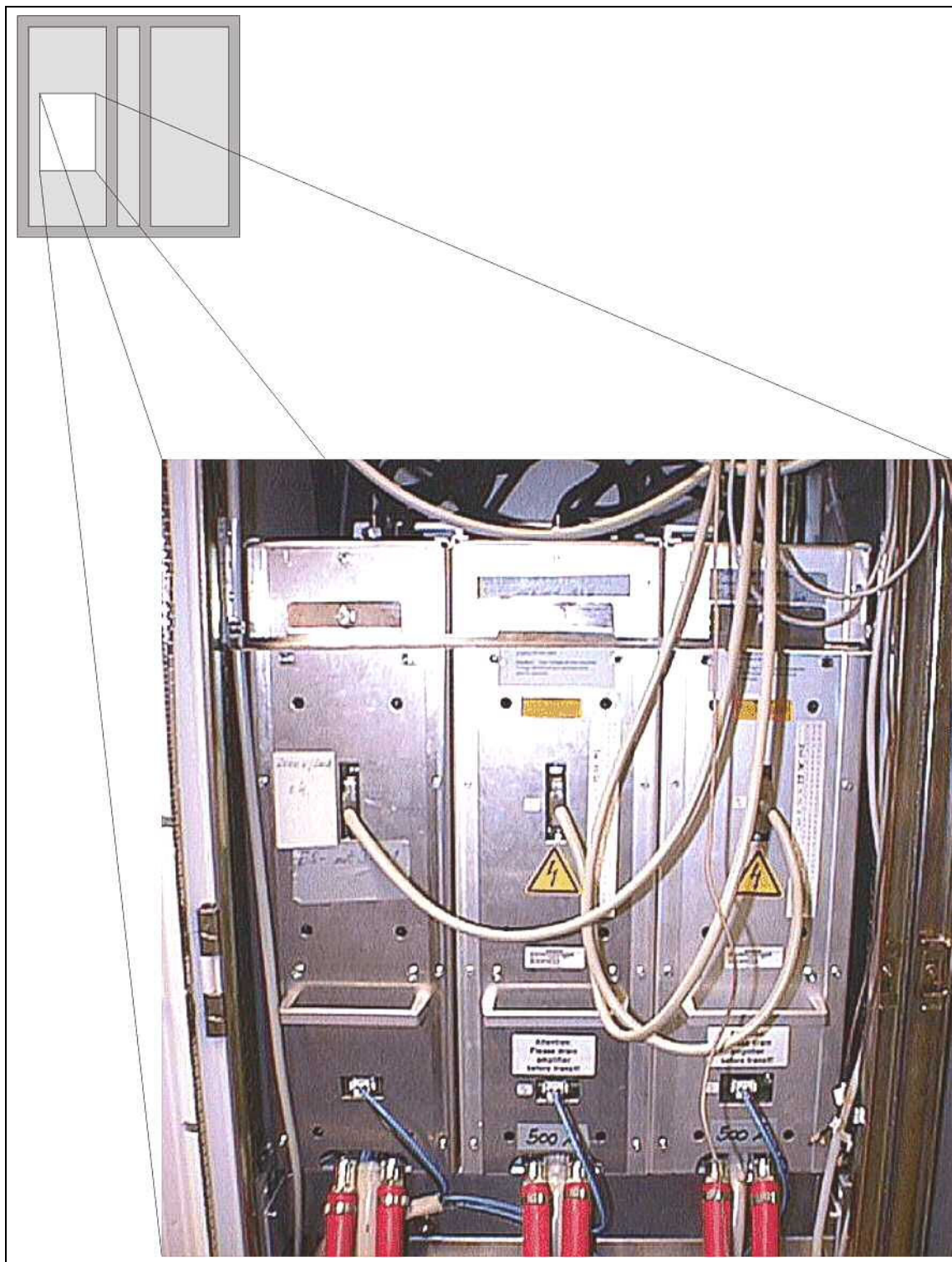
A normally closed contact of the relays K100...300 is used to inform the Service/CAN board about this error and the processor switches off the ON signal. The signal EPI\_P (Extreme Power Input) changes to the signal FASTOFF on the D17 board and lights up a red LED on the D16 board.

#### 4.2.1.2 Power-ON Acknowledge

If contactor K2 is activated (power-up finished), a normally open contact of K2 is used to switch on relay K3 on D31 via a cable interlock on D32. A normally open contact of K3 informs the processor. Service jumper X9 on D17 allows a bypass of this acknowledge signal.



Fig. 23: Power Stage Modules



## 5.1 Overview

Each of the three power stage modules consists of the following:

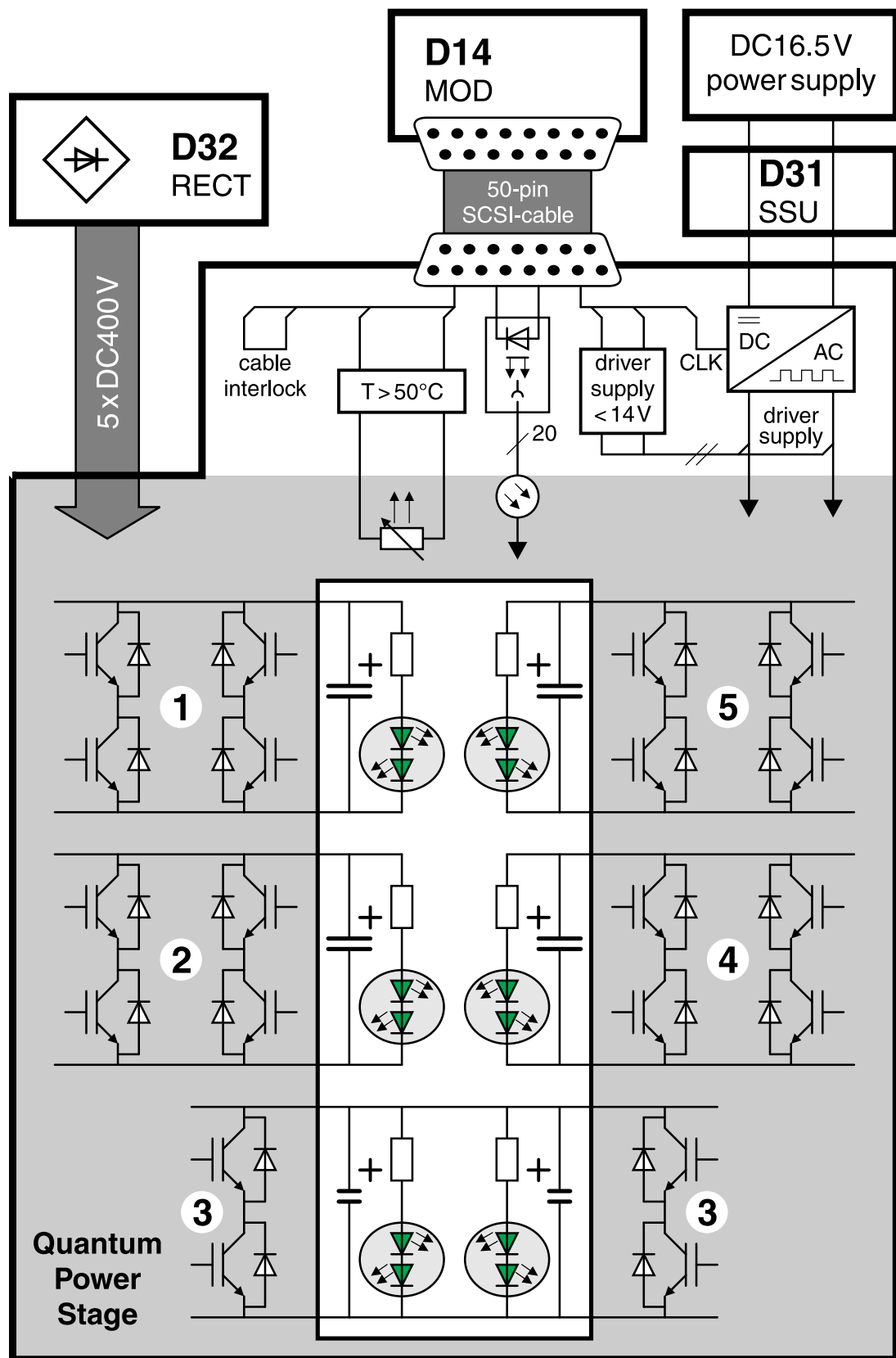
- 5 water cooled bridges (cascades) with 4 IGBT switches each (3 water cooled bridges in case of light).
- 5 times 400 VDC power stage supply.
- Integrated driver supply, capacitor block and filter units.
- Integrated opto-transmitter and opto-receiver for drive and monitoring signals.
- Maximum output 500 A and 2000 V



The power stage is a Field Replaceable Unit (FRU)!

## 5.2 Power Stage Module

Fig. 24: Power Stage Module



### 5.2.1 Connectors

- 5 times 400 VDC from the rectifier board D32
- 16.5 VDC supply from the GSSU power supply unit routed via the soft start unit D31.
- 50-pin SCSI cable for the drive signals coming from the modulator board D14.

### 5.2.2 LEDs

Each of the modules consists of 5 cascades (bridges) with its own 400 VDC supply. The presence of this supply voltage is indicated by six green double LED sets. The upper four show the supply of four cascades and the two on the bottom show the supply of the 5th cascade. The reason is simply layout design of the boards used.

## 5.3 Cascade Power Stage Switching

### 5.3.1 Function

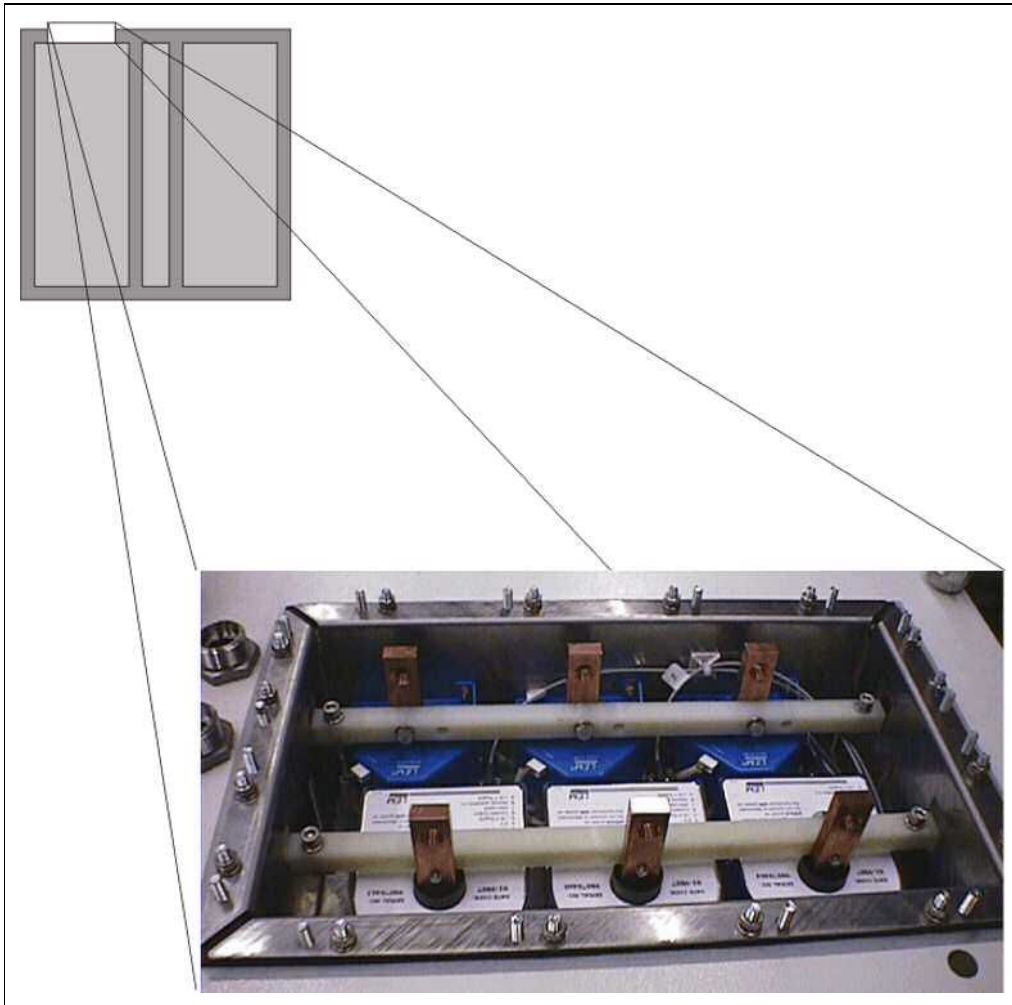
Each cascade consists of four switches. The switches are made from 6 IGBTs connected in parallel (insulated gate bipolar transistor) with respective free-wheeling diodes connected in anti-parallel.

*Fig. 25: Two switches containing 6 IGBTs each*



The switching principle of each cascade is identical to the Power Stage switching described in the MR Basics Gradient System documentation.

Fig. 26: Current Sensor Assembly



## 6.1 Overview

The Current Sensor Assembly consists of three LEM current sensors for monitoring functions and three DANFYSIK current sensors that supply the precise current value (actual value) for the regulation.

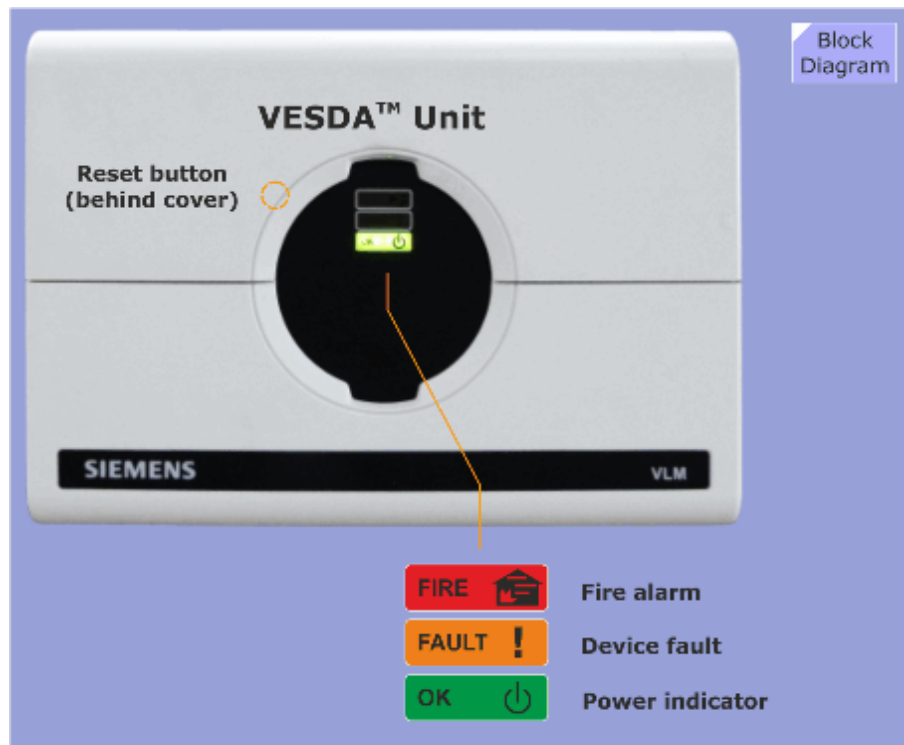
In addition, filter capacitors are installed in this unit.



The Current Sensor Assembly is a FRU!

## 7.1 Overview

Fig. 27: Gradient Supervision SIEMENS\_VLM



MAGNETOM Symphony a Tim Systems have an externally mounted **Gradient Supervision** unit called the **Siemens VLM** (a smoke detection product line) VESDA™ (product name). The unit incorporates a very sensitive laser-based light scatter detection system which can detect even small amounts of smoke particles emitted from overheated connections. The purpose of this unit is to provide continuous monitoring of all critical gradient connections.

Poorly fastened gradient cable connections have higher contact resistance which will lead to connector/cable overheating and damage and, in worst case, even fire.

The unit is integrated into the MR system. In case of a significant smoke particle concentration, a first relay is opened which will be processed by the AMC and an error message (evacuate patient) is posted to the monitor. One minute later, a second relay will open and the power distributor will be forced to switch-off the MR system to stand by mode.

The Gradient Supervision also provides a possibility to access stored data via a RS232 serial connection. With help of the optional software Medical VSC (contact HSC), the user can retrieve important data like progression of smoke particle concentration, alarm and warning history which may be helpful to analyze and understand fault scenarios.

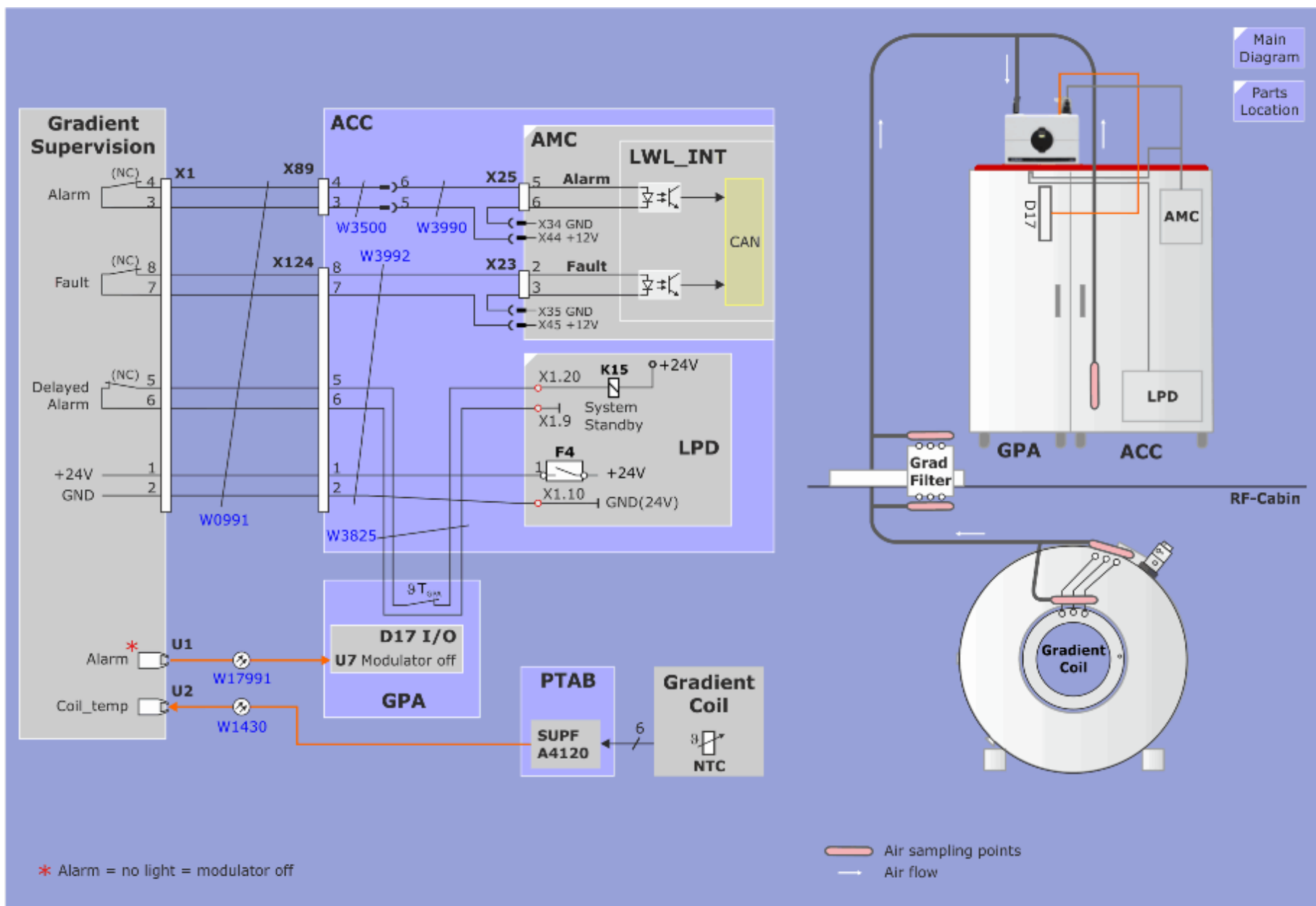


## 7.2 Function

Air is continuously drawn through a hose system which is mounted over all critical gradient connections and in the ACC cabinet. The air samples are fed through a filter. The filtering of dust and dirt from the air sample is important in order to keep the units optical detector surfaces free from contamination to ensure accurate readings and a long detector life.

Once filtered, the air sample is evaluated by a laser detector. If the sample contains smoke particles (e.g. emitted by overheated gradient connections) the laser beam is scattered which is detected by a highly sensitive optical receiver system.

Fig. 28: Gradient Supervision Block Diagram



## 7.2.1 Power Supply

The unit is powered with +24 VDC supplied by the 24/36 VDC power supply in the LPD (connector X124 at EPC roof). The supply voltage can be switched with circuit breaker F4.

## 7.2.2 Device Calibration

### 7.2.2.1 Warm-up

After an initial power-up, the yellow FAULT LED lights-up for 5 seconds and the unit runs through a 90 second warm-up cycle.

### 7.2.2.2 Reference Airflow Calibration

After initial installation of the unit it will be necessary to **calibrate** the nominal value of the reference air.



For correct calibration, the sample hose system must be fully attached. Also, close RF-room door to set standard air pressure condition in the RF-cabin.

To start the calibration:

- The unit must be powered-up for more than 90 seconds (= completed warm-up)
- Press the units' RESET button under the cover for 10 seconds until all 3 LEDs start flashing
- Release RESET button, all LEDs go off
- LEDs flash again, press RESET once more to confirm setting
  - » The current air flow through the device (which depends on the individual system configuration and hose length) is now **set to 100%**.
  - » Deviations larger than +/-30 % (+/-20 % at older VLMs) from the 100% nominal air flow (e.g. a blocked or disconnected sample hose) will result in a **device fault**.

## 7.2.3 Device Fault

### 7.2.3.1 Fault LED

In case of a malfunction the units **yellow FAULT LED** will light.

In fault mode, the supervisions' sensitivity is potentially affected, but the operation of the MR scanner is still possible. A constant light indicates a major fault, a flashing LED a minor problem.

The following list contains some possible causes for a device fault:

- **Wrong air flow** - more than +/-30 % (+/-20 % at older VLMs) deviation to the calibrated 100% reference air-flow value.
- **Excessive differential pressure** - more than 50 Pascal (0.5 mbar) between RF-cabin in-/ outside. This problem can be caused e.g. by an air condition system and will result

in the above mentioned air flow deviation of more than +/- 30 % (+/-20 % at older VLMs). A possible solution is, to route an equalization tube from the units exhaust back into the RF-cabin, thus equalizing the pressure difference seen by the VLM.

- **Defective laser scatter light detector**
- **Clogged VLM air filter**



The differential pressure between RF-cabin in-/ outside should not exceed 50 Pascal (0.5 mbar).

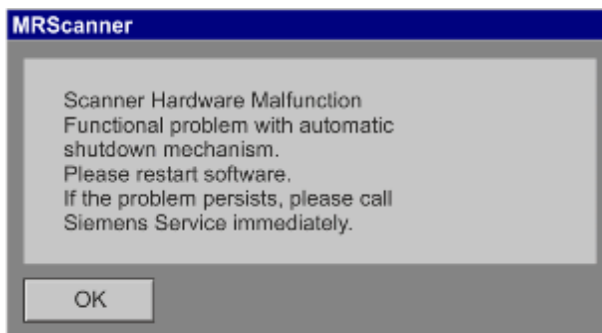
Otherwise, the VLM may report air flow device faults once the RF-room door is opened.

Ensure, the differential pressure is less than 50 Pascal (0.5 mbar) or install a pressure equalization tube from the units' exhaust back into the RF-cabin.

### 7.2.3.2 Device Fault pop-up

The customer is informed with a pop-up window.

Fig. 29: Device Fault pop-up



### 7.2.3.3 Device Fault Reset

A Device Fault will be **automatically reset** when the fault condition is not present anymore.

## 7.2.4 Fire Alarm

### 7.2.4.1 FIRE LED

In case smoke gases are detected within the monitored areas, the SIEMENS\_VLM sets the **red FIRE LED**.

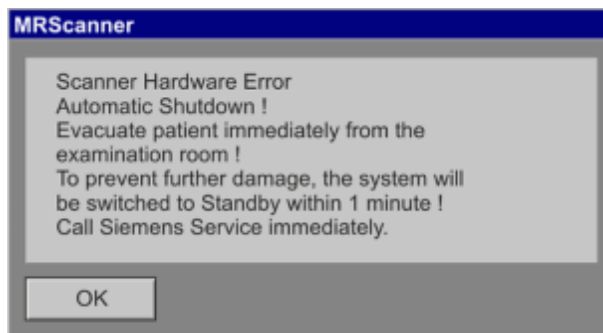
Fig. 30: Fire Alarm LED



#### 7.2.4.2 Alarm pop-up

The customer is informed about a fire alarm with a pop-up window.

Fig. 31: Fire Alarm pop-up



#### 7.2.4.3 Measurement Abort

The MR system is informed about the critical condition via the connection to the LWL\_INT (cable W0991). Consequently, a running measurement is aborted and the modulators of the GPA are blocked (via FOC W17991).

#### 7.2.4.4 Power Switch-off

After **one minute** the MR system is **switched to stand-by**. The signal is fed via X1/ pin 5, 6 VLM to X124/ pin 5, 6 to ACC roof and then in series to the GPA cabinet air temperature sensor.

#### 7.2.4.5 Fire Alarm Reset

A Fire Alarm condition is latched by the SIEMENS\_VLM to inhibit further system power switch-on attempts.

Once, the smoke concentration went below its alarm threshold, a CSE can reset the fault by pressing the units' **RESET button**. It can be found under the units' cover left to the indicator LEDs.

### 7.3 Jumpers

| Jumper | Pins | Setting | Location                                          | Description                                      |
|--------|------|---------|---------------------------------------------------|--------------------------------------------------|
| Config | 2-3  | closed  | under the units' cover, left of the FOC connector | no alarm = FOC light on<br>alarm = FOC light off |

## 7.4 Switches

| Switch | Location                                         | Description                                                                                                            |
|--------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| RESET  | under the units' cover, left of the control LEDs | <ul style="list-style-type: none"><li>■ Resets latched faults</li><li>■ Set-up button for device calibration</li></ul> |

## 7.5 LEDs

| LED               | ON             | FLASHING    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------|----------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OK<br>(green)     | Power on       | n.a.        | Detector is powered up and in normal operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| FAULT<br>(yellow) | Major Fault    | Minor Fault | <p>Call service, most common faults:</p> <ul style="list-style-type: none"> <li>■ Air flow deviation more than 30 % (20 % at older VLMs) from calibrated 100 % nominal flow, see (→ Device Fault / Page 59).</li> <li>■ Calibration of airflow required.</li> <li>■ Defective laser scatter light detector.</li> <li>■ Clogged VLM air filter.</li> </ul> <p>Note:<br/>Several additional, less probable faults exist. VLM faults can be further diagnosed with a service PC at the RS 232 serial interface and optional software "Medical VSC" (contact HSC).</p> |
| FIRE<br>(red)     | Smoke detected | n.a.        | <ul style="list-style-type: none"> <li>■ Alarm pop-up generated</li> <li>■ GPA modulator will be disabled</li> <li>■ MR-scanner will be switched off to stand-by 1 minute later</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |



## 7.6 Service Procedures

The function of the Gradient Supervision must be checked every 12 months during maintenance.



For details please refer to the Safety-Related Tests Procedure in the CB-DOC.

**Version 02:**

- » New section (→ Gradient Supervision / Page 56) added.

There are no Hazard IDs in this document.

- Restricted - All documents may only be used by authorized personnel for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

[healthcare.siemens.com/services](http://healthcare.siemens.com/services)

Siemens Healthineers Headquarters  
Siemens Healthcare GmbH  
Henkestr. 127  
91052 Erlangen  
Germany  
Telephone: +49 9131 84-0  
[siemens.com/healthineers](http://siemens.com/healthineers)

Print No.: M6-015.850.16.02.02 | Replaces: M6-015.850.16.01.02  
Doc. Gen. Date: 01.14 | Language: English  
© Siemens Healthcare GmbH, 2006

**[siemens.com/healthineers](http://siemens.com/healthineers)**